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Magnetostratigraphic constraints for Early Permian rocks of the southwestern Karoo Basin, South Africa

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ABSTRACT

We present results of rock and paleomagnetic magnetic analyses of a ~671 m-thick continuous vertical drill core (KZF-1) that intersect the lower Ecca Group (early-mid Permian) of the southwestern Karoo Basin, South Africa. Rock magnetic and optic microscopy experiments indicate monoclinic pyrrhotite and single-domain magnetite in the mudstones, shales, and siltstones as carriers of characteristic remanent magnetization (ChRM), which is likely a post-detrital remanent magnetization. Stepwise demagnetization and removal of low-coercivity and thermally less stable magnetizations reveal the preservation of a dual polarity ChRM in 90 samples. These are used to construct a magnetostratigraphic profile for the core that is dominantly reversed polarity with four short normal polarity subchrons. Correlation with published U–Pb SHRIMP and CA TIMS ages and the proposed composite reference section for the Early Permian allows us to propose an Artinskian (281 Ma) to Kungurian (276 Ma) age for the lower Ecca Group rocks. Our magnetostratigraphic profile can be tied in with published profiles from the mid to Upper Ecca Group to produce the first composite profile that spans all of the Ecca Group in the southwestern region of the Karoo Basin.

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1. Introduction

The Paleozoic- to Mesozoic Karoo Basin of South Africa developed due to crustal shortening caused by the subduction of the paleo-Pacific plate beneath Gondwana (Lock, 1978, 1980; De Wit and Ransome, 1992). This tectonic process accompanied by compression, collision, and terrane accretion along the southern margin of Gondwana formed other basins like the Paraná in South America, the Beacon in Antarctica and the Bowen in Australia. The Karoo Basin of South Africa is bordered in the south by the Cape Fold Belt; a small portion of the late Paleozoic Pan–Gondwanan fold-thrust belt (De Wit and Ransome, 1992; Veevers et al., 1994; (Visser and Praekelt, 1996)López-Gamundí and Rossello, 1998; Milani and De Wit, 2008). The well-established lithostratigraphic succession of the Karoo Supergroup reflects a changing paleoenvironment (Fig. 1) from the glaciomarine Dwyka Group at its base, followed by lacustrine or marine-deltaic Ecca Group strata, the fluvio-deltaic Beaufort Group, and the fluvial to aeolian lithologies of the Stormberg Group. The succession is capped by Jurassic-aged basaltic lavas of the Drakensberg Group (Veevers et al., 1994; Johnson et al., 1996, 1997, 2006). Contemporaneous with Permian marine incursion, and deposition of the Ecca Group, thrust loading of the lithosphere in the Cape Fold Belt caused a localized rapid deepening of the Karoo foreland basin (Cole, 1992). This led to the development of several depocenters of clastic deposition in the southern Karoo Basin (Catuneanu et al., 1998, 2002). The resulting clastic successions are highly asymmetrical, with the thickest sedimentary packages found in the southwest depocenters and rapidly thinning to the north-eastern part of the basin (Catuneanu et al., 2005; Rubidge et al., 2012). One such depocenter in the southwestern Karoo Basin is the Tanqua depocenter, which is bordered by the western Cedarberg branch of the Cape Fold Belt (Fig. 1) and hosts a ~ 6000 m-thick Permian succession of Ecca Group strata of the Karoo Supergroup.

The Ecca Group in the southwestern Karoo Basin is well described and understood in terms of its stratigraphy and depositional environment (De Beer, 1990; Cole, 1992; Veevers et al., 1994; Johnson et al., 1997; Visser, 1997; Catuneanu et al., 1998; Rubidge et al., 2000; Cole and Wipplinger, 2001; (Welman et al., 2001)Cole and Smith, 2008). Subdivision of the Ecca Group across the basin is based on lithostratigraphic analysis, but the temporal development and correlation of these units are more debatable due to conflicting views of the available biostratigraphic and chronostratigraphic records. Magnetostratigraphic studies have added additional datasets for understanding the temporal development of the Ecca Group. These datasets and their limitations are briefly summarized here.

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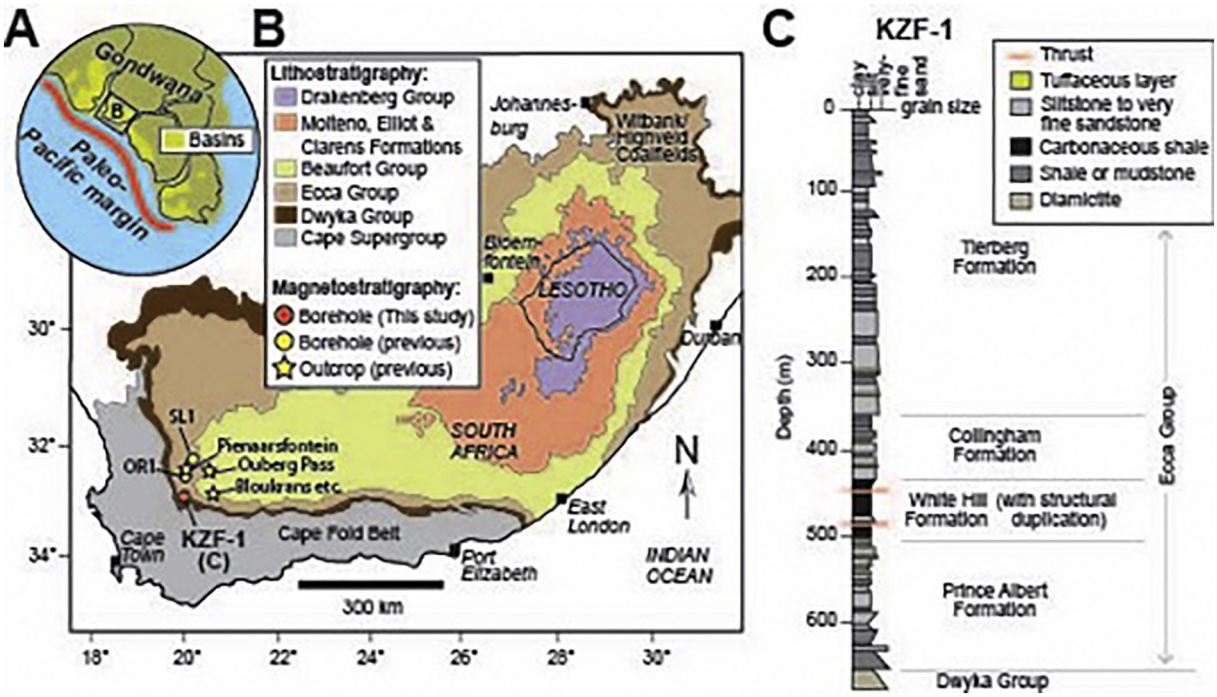


Fig. 1. Simplified geological map of the Karoo Basin showing the southwest bordered by the western Cedarberg branch of the Cape Fold Belt and location of the KZF-1 borehole. Stratigraphy of the Karoo Supergroup (modified after Johnson et al., 2006) and the investigated lower Ecca Group strata in the KZF-1 are shown Insets: (left) paleogeographic reconstruction of Gondwana during the Permian showing the Karoo Basin formed by subduction and accretionary tectonics along the southwestern margin (modified after De Wit and Ransome, 1992; Visser, 1992, Veevers et al., 1994; Faure and Cole, 1999); (right) cross-section showing the lithologies intersected in the KZF-1 drill core.

1.1. Palynostratigraphic record

The extensive records of well-preserved terrestrial fossil vertebrate biozones that establish temporal-spatial resolution for the fluvio-deltaic Beaufort Group (Keyser and Smith, 1979; Kitching, 1977; Rubidge, 1995; Smith and Keyser, 1995) is unfortunately not achievable for the lacustrine to marine Ecca Group. The Ecca Group is characterized by few age-diagnostic fossils and generally poorly preserved palynomorphs due to marine conditions (Visser, 1989, 1997), metamorphic effects of the Cape Fold Belt and thermal effects of dolerite intrusions that are related to extrusion of the Drakensberg Group (Barbolini et al., 2018). A handful of studies of pollen and spore taxa (Barbolini, 2014; Barbolini et al., 2016; Barbolini et al., 2018) allow correlation of southwestern Karoo Basin strata to that in the northern part of Karoo Basin where palynomorphs are better preserved (McLachlan and Anderson, 1975; Anderson, 1977; Aitken, 1998; Falcon et al., 1984; Ruckwied et al., 2014; Götz et al., 2018). However, there is some disagreement about such correlations. Most recently, Götz et al. (2018) presented a cross-basin correlation of the southwestern marine black shales of the Whitehill and Prince Albert formations with the north-eastern terrestrial Vryheid Formation No. 5 coal seam in the Witbank and Highveld coalfields based on Acritarch assemblages. They thus assigned an Early Roadian age (i.e., <272 Ma) to the Whitehill Formation. However, Cole and Barbolini (2019) use of Acritarchs as paleoenvironment indicators and they support a Kungurian age (283–272 Ma) for the Whitehill Formation based on lithostratigraphic correlations (Faure and Cole, 1999), palynological studies (Barbolini et al., 2018), isotope ages of zircon grains from correlative volcanic tuffs in the Aranos Basin of southern Namibia (Werner, 2006), and the stratigraphic-equivalent Irati Formation (Anderson and McLachlan, 1979) in the Paraná Basin of southern Brazil (Santos et al., 2006).

1.2. Chronostratigraphic record

There are no direct age constraints on the lower Ecca Group within the Tanqua depocenter, but volcanic ash beds from the Prince Albert Formation in the Laingsburg depocenter, located 50 km southeast, yielded U–Pb zircon ages of 288 ± 3 Ma and 289.6 ± 3.8 Ma, which is in agreement with ~290 Ma ages for the Dwyka-Ecca boundary in southern Namibia (Bangert et al., 1999).

Radiometric constraints from the middle Ecca Group in the southwestern part of the basin are conflicting. From the Tanqua depocenter, U–Pb ages of 275–254 Ma have been presented for the Laingsburg and Skoorsteenberg Formation of the middle Ecca Group strata (Fildani et al., 2007, 2009). These ages are disputed because of the implied Triassic age for the overlying Beaufort Group (Fildani et al., 2007, 2009; McKay et al., 2015, 2016). Belica et al. (2017) expatiated on the implication of these ages as a contradiction of existing paleontologic data. The most glaring of these contradictions is the placement of the geochronological Permian-Triassic boundary within the Upper Ecca Group, well below its biostratigraphically defined location in the Upper Beaufort Group (e.g., Botha and Smith, 2007; Ward et al., 2005). Belica et al. (2017) further reiterate that Permian-Triassic ages from the Ecca Group firstly, contradicts the ca. 268–264 Ma U-Pb SHRIMP age constraint proposed by Lanci et al. (2013) for interbedded volcanic ash beds from the Waterford Formation (upper Ecca Group) and fluvial sediments of the Abrahamskraal Formation (lower Beaufort Group); and secondly, disputes the biostratigraphic constraints of ~262–255 Ma for the overlying Adelaide Subgroup of the Beaufort Group (Rubidge et al., 2013). Belica et al. (2017) further presented a U–Pb age of 269.5 ± 1.2 Ma from the uppermost Tierberg Formation.

The younger ages of Fildani et al. (2007, 2009) were also questioned by Lanci et al. (2013) and Tohver et al. (2015). Lanci et al. (2013) suggested that younger zircons of Fildani et al. (2007, 2009) may be affected by undetected Pb-loss. Accordingly, Tohver et al. (2015)

showed that use of the youngest single concordant zircon grains for age assignments of ca. 254–274 Ma can be misleading. Tohver et al. (2015) re-assessed the data of Fildani et al. (2009) to identify ca. 265–275 Ma youngest zircon populations that are likely unaffected by Pb-loss.

1.3. Magnetostratigraphic record

Magnetostratigraphy is a valuable tool for both local and global correlations (Opdyke and Channell, 1996; McElhinny and McFadden, 2000; Ogg et al., 2016). Globally correlative magnetostratigraphic patterns within Karoo sedimentary rocks span the late Triassic to early Jurassic Upper Elliot Formation, the biostratigraphic Permian-Triassic boundary in the Beaufort Group, as well as the Permian mid to Upper Ecca Group and its contact interval with the Beaufort Group (Kirschvink and Ward, 1998; De Kock and Kirschvink, 2004; Ward et al., 2005; Lanci et al., 2013; Sciscio et al., 2017; Tohver et al., 2015; Belica et al., 2017). For constraining the timing of Ecca Group deposition the most important constraint comes from the magnetostratigraphies of Lanci et al. (2013) and Tohver et al. (2015), who place the Ecca-Beaufort Group contact and Lowermost Beaufort Group above the Kiaman Reversed Superchron – an important stratigraphic marker and long polarity interval that lasted some 50 million years from the late Carboniferous to the late Permian from ~265 to 318 Ma (Irving and Parry, 1963; Opdyke et al., 2000; Steiner, 2006; Hounslow and Balabanov, 2018).

In a test of published U–Pb age data, Belica et al. (2017) compared a 1.5 km thick magnetostratigraphic section covering the mid to Upper Ecca Group strata in the Tanqua depocenter with the composite reference section *proposed* by Hounslow and Balabanov (2018). Based on predominantly reversed polarity that suggests deposition during the Kiaman Reversed Superchron, Belica et al. (2017) re-established the location of Permian-Triassic boundary within the overlying Beaufort Group in support of a ~ 290–265 Ma age for the Ecca Group. The correlations of two poorly defined normal subchrons (each represented only by a single sample) in the Skoorsteenberg and Waterford formations (see Fig. 2) with normal subchrons in the section proposed by Hounslow and Balabanov (2018) is debatable. There is large uncertainty surrounding subchron definitions that are based on single samples. The age constraint of ca. 290–265 Ma proposed by Belica et al. (2017) for the Ecca Group rocks thus requires further testing. Moreover, the magnetostratigraphic constraints of ~276 Ma and ~ 281 Ma proposed by Belica et al. (2017) for the Kookfontein and Skoorsteenberg formations respectively are inconsistent with their own maximum U–Pb SHRIMP age constraint of 269.5 ± 1.2 Ma obtained from the top of the underlying Tierberg Formation (Fig. 2). It is also important to note that the study by Belica et al. (2017) is restricted to the Upper Ecca Group strata and includes only the top ~400 m of the Tierberg Formation in the Tanqua depocenter (Fig. 2), and not its entire ~750 m-thickness (Wickens, 1994). The magnetostratigraphic pattern for the lower part of the Tierberg Formation and the underlying Collingham, Whitehill and Prince Albert formations of the Ecca Group remains undefined (Fig. 2). Here we present a continuous 671 m-thick magnetostratigraphic section and provide rock magnetic analyses of samples from the KFZ-1 drill core covering the lower Ecca Group strata.

2. Geological setting of KZF-1

KZF-1 is located on the farm Zandfontein 89 in the Witzenberg district (coordinates S32 50′ 30.43″ E19 49′ 33.02″) of the Western Cape Province of South Africa (Fig. 1). This location was selected south of the southern limit of dolerites belonging to the Karoo Large Igneous Province and near the lower contact of the Ecca Group with the Dwyka Group. This drill site falls within the Cape syntaxis; an oroclinal bend formed by continued dextral shearing during the Late Jurassic to

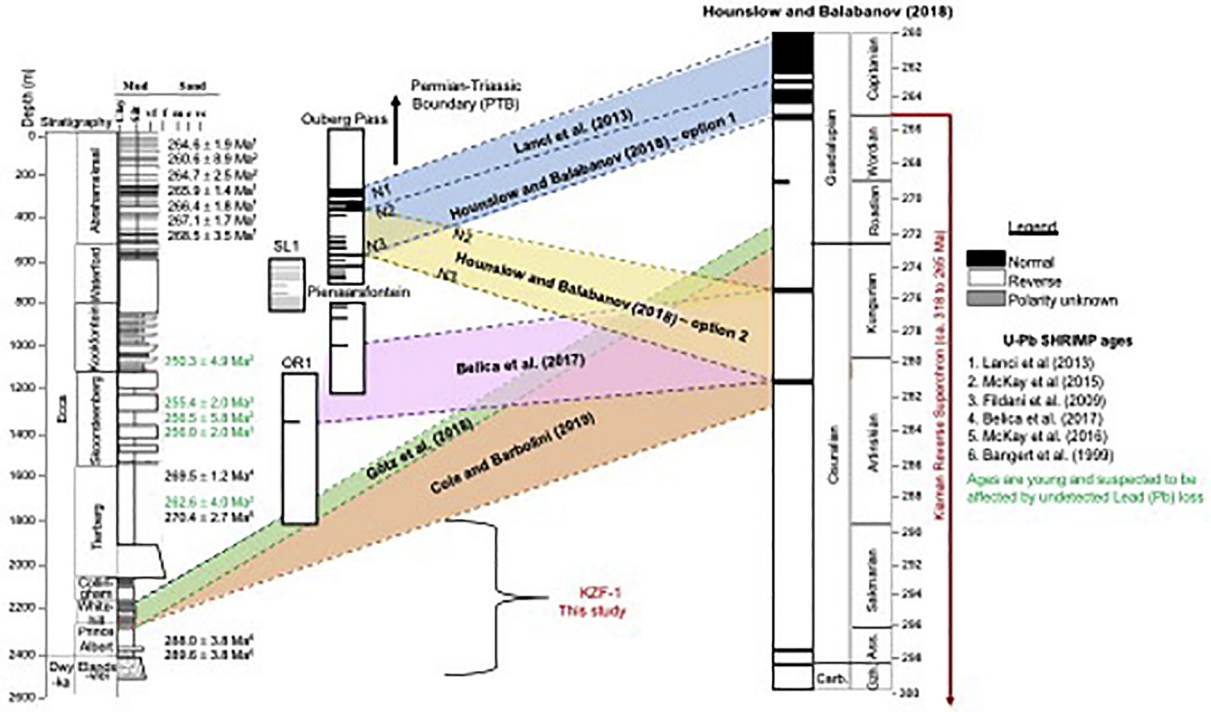


Fig. 2. Lithostratigraphy and magnetostratigraphic sections for the mid to Upper Ecca Group (Belica et al., 2017) and lower Abrahamskraal Formation of the Beaufort Group (Lanci et al., 2013). Conflicting palynostratigraphic and magnetostratigraphic ages explained in text are shown. U–Pb ages in green are considered to be young and suspected to be affected by undetected lead loss. The lower Ecca strata covered by the KZF-1 (this study) is indicated. Lithostratigraphy after Johnson et al. (2006); proposed composite reference section after Hounslow and Balabanov (2018); Carb. = Carboniferous; Gzh. = Ghezelian; Ass. = Asselian.

Early Cretaceous Epochs that splits the Cedarberg and Swartzberg branches of the Cape Fold Belt (Johnston, 2000).

KZF-1 core provides a continuous 671 m section devoid of any dolerite intrusion that intersects four formations of the deepwater lower Ecca Group strata (namely the mid to lower Tierberg; Collingham; Whitehill and Prince Albert formations) and ultimately, the uppermost layers of the glacial Dwyka Group. The top of KZF-1 is within the middle of the Tierberg Formation and composed of 338.62 m thickness of succession that shows three distinct lithofacies, namely a topmost layer of massive dark grey to grey shale with mudstone interbeds; a middle lithofacies consisting of a finning-upward alternation of parallel laminated grey siltstone to very fine-grained sandstone (Fig. 3i), thin massive shale interbeds and scattered calcareous mudstone laminae; and at the bottom, an upward-thinning rhythmite consisting of grey to light grey calcareous siltstone to very fine-grained sandstone layers (Fig. 3h), thin (5–10 mm thick) light olive grey coloured massive shale beds with mud clast bearing mudstone laminae inclusions.

The light-coloured laminae at the bottom of the Tierberg Formation in our section suggest a neritic and diatomaceous mud belt reflecting strong upwelling from deeper layers in the Karoo Basin while the darker laminae indicate reduced upwelling and incursion of offshore water in the slope margin. Also observed in the lithofacies are limited bioturbation in the siltstone laminae at the upper portion of the lithofacies, which suggests depletion of dissolved oxygen in a bottom water setting and occasional soft-sediment deformation within the siltstone beds suggesting internal deformation due to rapid influx of sediments exceeding accommodation space (Catuneanu et al., 2002). The general fining-upwards and rhythmic pattern of the Tierberg Formation are indicative of a turbidite unit formed by the influx of sediments into the southwest Karoo Basin initially by low-density turbidity currents to a transitional flow consisting of high mud content at the distal regions of the turbidite system (Talling et al., 2012; Sumner et al., 2009; Boulesteix et al., 2019).

The Collingham Formation is intersected between 420.24 and 347.51 m and is recognised as an 81.84 m-thick unit of upward-fining, alternating beds of dark grey calcareous and laminated siltstone interbedded with thin calcareous yellowish or light-greyish mudstone (Fig. 3g), and thin partially cemented massive grey shale lenses mostly concentrated towards the uppermost part of the formation and numerous tuffaceous layers throughout the unit. The light grey mudstone interbeds (observed at the upper part of this facies association (top 40 m), have abundant light grey calcite and phosphatic concretions and a characteristic soapy-clay feel identified as K-bentonite rich tuff layers (Visser, 1993; Viljoen, 1994). Also observed at the uppermost part of the Collingham Formation is a 4.95 m-thick condensed organic-rich, radioactive shale layer (Fig. 3f) with abundant vertical burrowing (bioturbation index 5) and mud clasts. This shale layer is interpreted as a marine condensed section and is consistent with a prolonged period of sediment starvation on the basin floor restricted to deposition of only very fine-grained, siliciclastic, organic and biogenic particles (Slatt, 2006).

The core records duplication of the Whitehill Formation caused by two tectonically displaced layers of the underlying Prince Albert Formation. A thickness of 19.5 m of carbonaceous mudstone was initially intersected from 439.95–420.46 m-depth, followed by further intersections of 36 m thickness from 443.30–479.55 m-depth and a 9 m thickness from 489.15–498.45 m-depth. Despite the difficulty in determining the true thickness of the Whitehill Formation given the duplication, the Whitehill Formation is easily distinguished from the underlying Prince Albert Formation, and a likely 64.5 m-thickness of highly fissile carbonaceous and pyrite-rich shale and black mudstone (Fig. 3d) is suggested to make up this unit. A highly brecciated massive

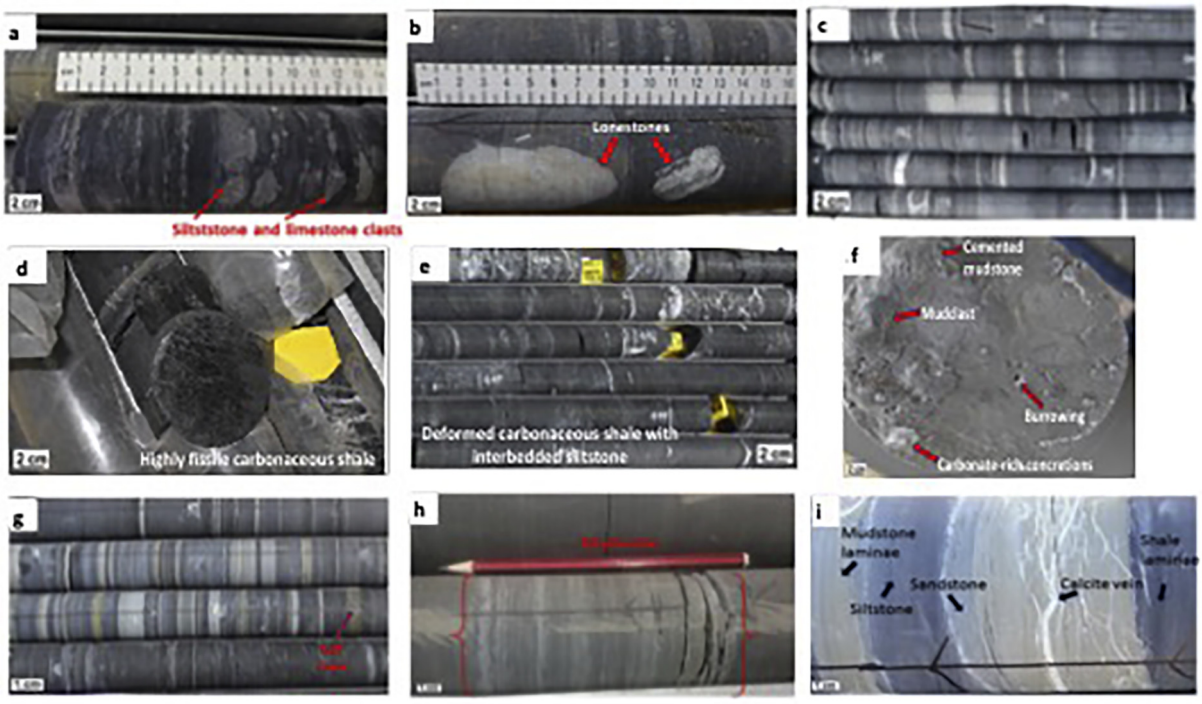


Fig. 3. Core photographs of the lower Ecca Group strata in KZF-1; a) diamictite from the Dwyka Group at 666.54 m depth; b) laminated mudstone with isolated lonestones and shale interbeds from the lowermost portion of the Prince Albert Formation at 648.23 m depth; c) calcareous and laminated siltstone to very fine-grained sandstone bed with thinly laminated silty mudstone and thick grey shale interbeds from the middle to uppermost portion of the Prince Albert Formation at 528–534 m depth; d-e) highly fissile and deformed carbonaceous shale of the Whitehill Formation between 449 and 443 m depth; f-g) partially cemented mudstone with carbonate-rich concretions and mud clasts at 362.13 m depth, and alternating beds of dark grey calcareous laminated siltstone with thin calcareous yellowish to light-grey mudstone interbeds (with K-bentonite rich tuffaceous layers) and thin shale lenses from the Collingham Formation at 421–423 m depth; h-i) rhythmites of calcareous siltstone to very fine-grained sandstone interbedded with thin massive shale laminae at 225 m depth, and at 150 m depth from the lowermost to middle portion of the Tierberg Formation.

grey shale zone interbedded with deformed grey siltstone in a repeated pattern is observed in the middle of the lithofacies and at the base of the lithofacies is a light grey to grey steeply dipping black siltstone laminae with soft-sediment deformation, abundant quartz veins and fine-grained pyrite lenses. The presence of lamination with thin silty laminae with soft-sediment deformation within this facies association suggests deposition by low-energy cohesive debris flows or a mixture of laminar and turbulent flows (Boulesteix et al., 2019). The presence of abundant pyrite lenses and nodules (of up to 2 cm in diameter) and the lack of trace fossils indicate a severely reducing, euxinic and anoxic bottom-water environment of very low sediment accumulation. The presence of black shale and the indication of strongly anoxic marine conditions suggests low-energy sediment fallout in a deep-water setting during a sea-level highstand (Visser and Young, 1990; Visser, 1993; Nolte et al., 2019) of near-freezing bottom water conditions (Schulz et al., 2018).

The Prince Albert Formation is recognised as an unduplicated 167.97 m-thick alternation of grey massive shale beds and grey to dark grey coloured mud clast-bearing silty mudstone layers with scattered tuffaceous horizons (Fig. 3c). The upper 77.51 m of this facies is composed of ~50% alternation of siltstone to very fine-grained light grey to grey sandstone, 20% light to very light grey-coloured brecciated and deformed dark grey shale beds and 30% of carbonaceous shale beds consisting of moderate amounts of detrital grains. At the base of this lithofacies is a 35.6 m-thick alternating bed of thin calcareous and laminated greenish-grey coloured siltstone beds with thin beds of mud clast-bearing silty mudstone. At the Lowermost portion of the Prince Albert Formation, where it borders the underlying Elandsvlei Formation of the Dwyka Group is a ~ 39.91 m-thick light grey to grey mudstone layer with scattered coarse-grained to granule-sized (up to 4 mm) siltstone, sandstone and quartzite lonestones (Fig. 3b), carbon-rich plant fragments, quartz and mica grains and pale yellowish-brown coloured mud clast. The post-depositional structures identified within this lithofacies association are very fine-grained pyrite lenses, numerous steeply dipping quartz veins, soft-sediment deformation, abundant 2-3 cm-thick phosphatic concretions, mud clasts and a high degree of bioturbation (bioturbation index of 5).

The thin-bedded and planar parallel silty mudstone and shale laminae in this lithofacies suggest periodic fine-grained suspension settle-out of hemipelagic mud in quiet water or low energy environment. The influx of coarser siltstone and fine-grained sandstone beds suggests an increase in depositional energy caused by periodic storm-induced currents and deposition of detrital nepheloid layers in a semi-restricted (dysoxic to anoxic) offshore marine environment. The intense burrowing activities observed in the mudstone suggest an oxygenated sea-floor basin environment undergoing low sediment accumulation with intermittent low-frequency flow events (Rickard, 2012; D'Hondt et al., 2015). Lastly, the presence of abundant phosphatic concretions, mud clasts and the isolated lonestones (large clasts) enclosed in mud matrix at the base of this lithofacies is indicative of the existence of glaciers and deposition in a fresh glacial meltwater lake (Herbert and Compton, 2007) deposited by low-energy cohesive debris flow caused by the collective melting of the ice sheet from the underlying glacial Dwyka Group.

The Elandsvlei Formation of the Dwyka Group represents Gondwanan glaciation in the southern part of the main Karoo Basin of South Africa (Visser, 1989; Scheffler et al., 2006; Herbert and Compton, 2007; Schulz et al., 2018). This lithofacies marking the end of KZF-1 is intersected from 669.72–657.12 m depth and consists of a 12.6 m-thick poorly sorted diamictite bed with silty mudstone matrix-supported coarse-grained to pebble-sized (10–20 mm in size) rare sandstone, quartzite, siltstone and limestone clasts (Fig. 3a). The Dwyka Group diamictite represent meltwater flow deposition carrying loads of debris and the occurrence of vertical deposition in the water column rather than by water current.

3. Methods

3.1. Orientation of drill core

During drilling of KZF-1, relative core orientation was achieved by a geological manager, who carefully matched and refitted retrieved cores. Fractures or breaks where the spinning of the core within the core barrel occurred before retrieval, and where continuous fitting of the core could not be achieved were noted. This systematic approach allowed the core to be partitioned into several "intact" intervals without relative rotations. A reference orientation line was then ascribed on continuous core intervals. Reference lines were drawn parallel to the core axis to show the stratigraphically up direction of the core. Reference lines were drawn as to best align across the apexes of sinusoidal intersections of bedding planes (i.e., along the up-dip azimuth). These reference lines and the core could then be re-oriented based on the radial distance between the up-dip azimuth of bedding features and the reference line, the core diameter, and the bedding or structural data in the cores acquired by geophysical survey post drilling. The orientation of bedding planes and other structural features such as veins in the cores are constrained by both a dipmeter downhole survey as well as televiewer imaging of the hole. This technique is similar to that used by several other studies (Jarrard et al., 2001; Paulsen et al., 2002; De Kock et al., 2009; Shigematsu et al., 2014). As an example: At around 440 m depth a brecciated shale bed was intersected. The up-dip azimuth of the bed is observed as an apex in the curvature of the bed intersection (Fig. 4), and it is observed to be offset to the right of the inscribed reference line marking the relative orientation of an interval of the continuously fitted retrieved core. The shale bed (and its up-dip azimuth) is also identifiable on the televiewer image of the borehole wall obtained during a downhole survey after drilling was completed. The actual orientation of the up-dip azimuth can be read off as southerly (Fig. 4). The radial distance between the apex of the vein intersection and the reference line combined with the diameter of the core allows determination of the orientation of the reference line.

3.2. Sampling

KZF-1 drill core, which is curated at the Council for Geosciences National Core Library at Donkerhoek (Pretoria East) was logged and described. Samples for paleomagnetic analyses (i.e., $n = 225$) were collected at ~3 m intervals and prepared by splitting the cores into halves along relative orientation reference lines (see the previous section). Shale, siltstone and very fine-grained sandstone interbedded with mudstone were preferentially selected where possible because finer-grained lithologies are likely to be superior magnetic recorders (Butler, 1992). The right-hand side halves of sampled core pieces were retained for plugging of paleomagnetic specimens, while left-hand side halves were returned to the National Core Library for curation. Plugs of 2.5 cm outside diameter were drilled from core halves perpendicular to the core axis using a benchtop drill press. Plugs were labelled with their depth as well as azimuth on their upper margins with a non-magnetic and heat resistant white ink marker. Plug azimuths are equal to the calculated orientation of the reference line minus 90°. The plunges of plug azimuthal marks were taken to be near horizontal given the close to vertical nature and minimal vertical deviation of the borehole. Plugs were trimmed into 1–1.5 cm high cylindrical specimens. Azimuthal orientation allows the specimen to be restored to in-situ positions, while the bedding for each specimen allows restoration to paleohorizontal. Bedding restorations were generally very minimal to negligible given the average 4° easterly dip of strata (see Fig. 4).

3.3. Paleomagnetic measurements

All paleomagnetic measurements were obtained using a 2G Enterprises model 755 superconducting rock magnetometer housed in a

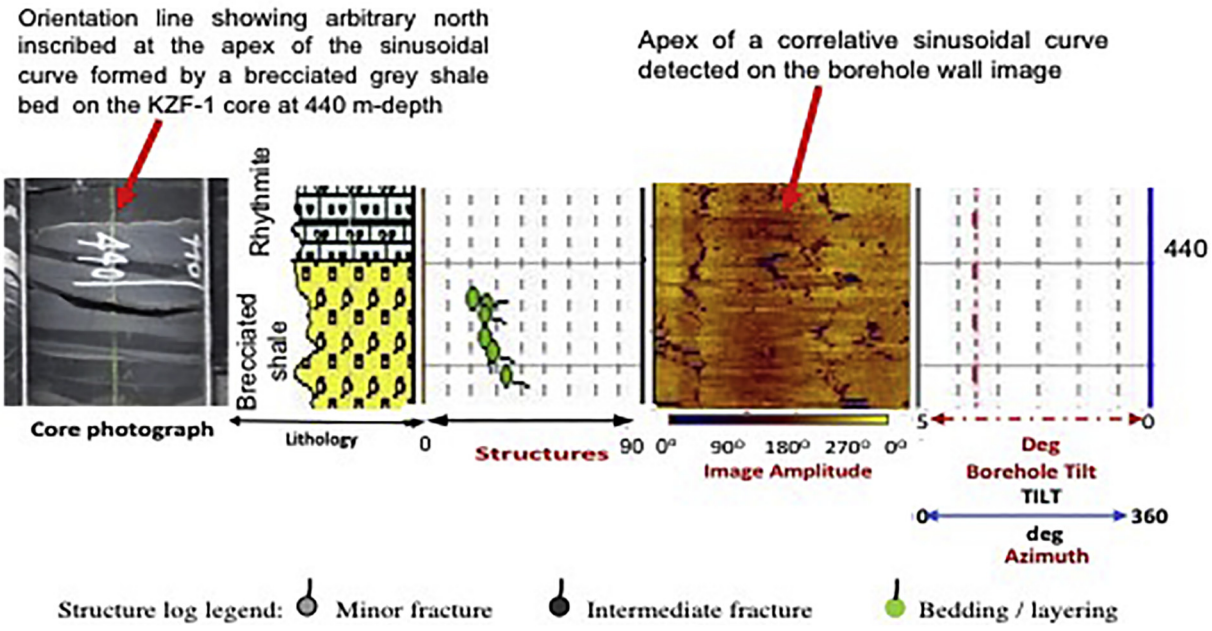


Fig. 4. Structure log from the borehole televiewer showing an example of matching features on drill core photograph and borehole wall image used to reorient KZF-1.

double-layer, mu-metal shielded room with ambient fields of <200nT of the Paleomagnetism Laboratory at the California Institute of Technology (Caltech). Specimen and the sample holder were cleaned with 95% ethanol and blown with high-pressure air passed through a 0.2-μm particulate filter to remove residual particles before commencing with measurements. The specimen was measured in their natural state to obtain their natural remanent magnetization (NRM). Hereafter, the specimen was subjected to low-temperature demagnetization cycles by dunking in liquid nitrogen for 20–30 min to remove any vicious components that might be carried by multi-domain magnetite. This was followed by 3 steps of low-intensity progressive Alternating Field (AF) demagnetization up to 7mT to remove any soft magnetic components acquired during transport or preparation. This was subsequently followed by thermal demagnetization in a magnetically shielded oven (< 25nT residual field) under a gentle flow of nitrogen gas to prevent the ferromagnetic minerals from being oxidized and recording new components while at high temperature. Stepwise thermal demagnetization was achieved by heating the specimen from 100 °C to 580 °C in increments of 20 °C. Demagnetization was stopped when the specimen magnetization dropped below the noise level of the sample handler or if specimen started displaying unstable behaviour.

Magnetic components recorded in the specimen were visualized and quantified via principal component analysis (Kirschvink, 1980) using PaleoMag 3.1 (Jones, 2002) and Multiplot (Lushkin et al., 2018) with individual components calculated from three or more points. These fits were only used in subsequent analyses if their Mean Angular Deviation (MAD) values were ≤ 15°. Least square lines were anchored to the origin for components that were observed as trending towards the origin or forming a stable endpoint cluster before displaying unstable behaviour. The arc method of McFadden and McElhinny (1988) was used to calculate planes and arc-constraints for samples that do not reach stable endpoints. This involved isolating the primary component of magnetization, first by determining the final iterative directions of planes (which is the great circle path of demagnetization trajectories of the samples) along with the converging arc constraints in Paleomag 3.1. Then the obtained directions for the planes are combined with the directions for the lines segments endpoints for the stratigraphic polarity interpretation and pole calculation.

3.4. Rock magnetic measurements

The magnetic mineralogy of representative lithologies from KZF-1 was determined in a suite of non-destructive (with no heat applied) rock magnetic experiments at Caltech. The rock magnetic experiments, which are described in Kirschvink et al. (2008), Kirschvink et al., 2015), include 3-axis Alternating-Field (AF) progressive demagnetization of the NRM of samples in peak fields up to 80 mT and acquisition and demagnetization of isothermal and anhysteretic remanent magnetizations (IRM and ARM). IRM/ARM coercivity spectral analysis allows identification of the ferromagnetic minerals present and constrains the destructive field value range (mT) and saturation peak pulse fields (mT). An ARM modified version of the Lowrie-Fuller test (Johnson et al., 1975) was used to identify the magnetic domain states of the ferromagnetic minerals. This required progressive acquisition of ARM by the sample through the application of a small range DC biasing field of 0–1 mT parallel to a slowly decreasing alternating field (AF) with a peak amplitude of 100 mT. ARM acquisition of samples (after Cisowski, 1981) in the coercivity band of 100 mT was used to determine the interparticle interaction characteristics. This involved progressive AF demagnetization of the ARMs, a stronger field with increasing peak-fields. Here the ARM substitutes for a weak-field Thermal Remanent Magnetization for Single-Domain (SD) and Pseudo-Single-Domain (PSD) particles. NRM origin was constrained using the Fuller test (Fuller et al., 1988) that compare the intensity of the NRM remaining during AF demagnetization with that of the IRM. All rock magnetic experiments utilized matRockmag MATLAB scripts available at https://github.com/bobkopp/matRockmag.

3.5. Optical microscopy

Thin section slides were prepared from representative samples at the University of Johannesburg and the prepared slides were examined with a Zeiss Axioplan 2 compound light microscope. Photographs were taken under 50–200 μm magnification. A colour view soft imaging system, AxioCam HRC digital camera, and AnalysisTM software were used for digital photography.

Table 1
Rock magnetism results of specimens from the lower Ecca Group strata. DG = Dark grey; G = Grey; LG = Light grey; GG = Greenish grey; OG = Olive grey; SD = Single-Domain; DRM = Depositional Remanent Magnetization; and pDRM = post-Depositional Remanent Magnetization.

Formation (Lithology)	Sample depth (m)	Destructive field range (mT)	Peak saturation (mT)	ARM acquisition (at 100 mT)	Lowrie–Fuller test	Fuller NRM
Tierberg						
(DG shale)	5.41	75–80	500	Less interacting	SD	DRM/pDRM
(G/LG Siltstone)	108.72	45–50	300	Mixture	SD	DRM/pDRM
(LG Rhythmite)	318.85	40–45	300	Mixture	SD	DRM/pDRM
(LG Siltstone)	330.23	45–50	300	Mixture	SD	DRM/pDRM
Collingham						
(LG Silty mudstone)	373.44	45–50	400	Mixture	SD	DRM/pDRM
Whitehill						
(GG Silty shale)	492.31	40–45	400	Mixture	SD	DRM/pDRM
(GG Shale)	499.73	35–40	400	Less interacting	SD	DRM/pDRM
Prince Albert						
(OG Siltstone) (LG Silty mudstone)	519.35	40–45	300	Less interacting	SD	DRM/pDRM
Dwyka Group (LG Diamictite)	656.22	45–50	400	Less interacting	SD	DRM/pDRM

4. Results

4.1. Rock magnetism

The IRM/ARM coercivity spectra analysis show that samples fall into one of two categories. About 60% of the samples from KZF-1 are mudstone and shale that show low destructive field range (coercivity) of 40–45 mT (Table 1). The remaining 40% are siltstone, to very fine-grained sandstone samples (particularly samples from the middle section of the Tierberg and Collingham formations) show higher coercivity values of between 45 and 50 mT (Table 1). An exception is seen in a shale sample collected from the top of KZF-1 (from the Tierberg Formation) showing a destructive field range between 70 and 80 mT (Table 1). A smaller saturation peak of 300 mT was measured for the siltstone samples (left plot in Fig. 5a), whereas the shale samples showed a clear approach to a high saturation remanent magnetization of 400–500 mT (right plot in Fig. 5a). Based on the coercivity and saturation remanent magnetization ranges recorded by Peters and Dekkers (2003), the siltstones and very fine-grained sandstone likely contain a small coercivity antiferromagnetic mineral-like magnetite, whereas the high saturation peak values recorded by the shale and mudstone samples suggest the presence of pyrrhotite.

The IRM/ARM demagnetization curves appear symmetrical and the IRM acquisition line of slope (RHS curve) is gentler and show saturation peak of 300 mT in the siltstone and very fine-grained sandstone (rhythmite) samples of the Tierberg Formation (Fig. 5a). Whereas for the Whitehill Formation shale samples, the IRM/ARM demagnetization curve is asymmetrical and the IRM acquisition line of the slope is steeper and shows a saturation peak of 400–500 mT (RHS curve) (Fig. 5a). The point of decay of the antiferromagnetic mineral (LHS curve) in the siltstone samples is 100 mT, while 80 mT was recorded for the shale samples.

The results for the ARM version of the Lowrie–Fuller test (Johnson et al., 1975) shows that in all samples, the ARM progressive AF demagnetization curve lies above that of IRM (Fig. 5b). This is indicative of Single-Domain (SD) or Pseudo-Single-Domain (PSD) magnetic grains (McElhinny and McFadden, 2000; Kirschvink et al., 2015). The relative separation between the two curves indicates that magnetic clumping occurred when exposed to the applied field.

Our Fuller test of NRM origin (Fig. 5c) shows NRM values that are about three orders of magnitude less than the corresponding IRM level and two orders of magnitude below the ARM values. This suggests that the NRM signal is a Depositional or post-depositional Remanent Magnetization (DRM or pDRM), rather than a Chemical Remanent Magnetization (CRM) or Thermal Remanent Magnetization (TRM; McElhinny and McFadden, 2000).

For the interparticle interaction characteristics as determined by ARM acquisition in the coercivity band of 100 mT (Fig. 5d), the results for all the siltstone samples plot halfway between the chiton tooth and magnetotactic bacterial curves (left plot in Fig. 5d), indicating the presence of a mixture of non-interacting and interacting particles. In contrast, shale samples from 499.73 m depth (right plot in Fig. 5d) and 5.41 m (see Table 1) follow a trajectory indicating slightly less affinity for inter-particle interactions similar to a slightly collapsed magnetotactic bacterial reference curve calibrated by Kobayashi et al. (2006).

4.2. Petrographic analysis

Framboidal and euhedral microcrystalline pyrite microcrystals were observed along bedding surface in the Whitehill Formation (Fig. 6a) and as scattered euhedral aggregate clusters in the Collingham Formation (Fig. 6b). The observed pyrite crystals serve as a paleoenvironmental indicator of lower temperature conditions (Aidona et al., 2001; Ohfuji and Rickard, 2005), and the morphology is indicative of formation due to direct precipitation from solution in a euxinic and reducing environment (Liu et al., 2019; Sawlowicz, 1993). However, the morphology and preserved framboidal structure of the pyrite crystals do not show any replacement by any reactive iron sulphides. There was also no observed clustering of magnetite around the pyrite crystals nor oxidized rims around the magnetite grains to signify subsequent sulphidation of the newly formed magnetite to pyrrhotite.

Thin sections from the lower portion of the core showed agglomeration of fine-grained euhedral magnetite grains (Figs. 6c and e), typical of magnetofossils (Kopp and Kirschvink, 2008; Egli et al., 2010; Roberts et al., 2012; Chang et al., 2014). This morphology has been interpreted as a biosignature for thermophilic iron-reducing bacteria known to enzymatically reduce ferric iron and fasten to the oxidation process of organic matter (Taylor et al., 2001; Li, 2012). An observed pyrrhotite grain in Fig. 6d appears to have a pseudo-hexagonal structure (most likely monoclinic) and is partly altered with the progressive development of new grains (greyer than the original pyrrhotite) possibly pyrite at the cleavage along cracks and grain boundaries. Evidence of co-existing pyrrhotite and small-sized magnetite grains of bacterial magnetosomes

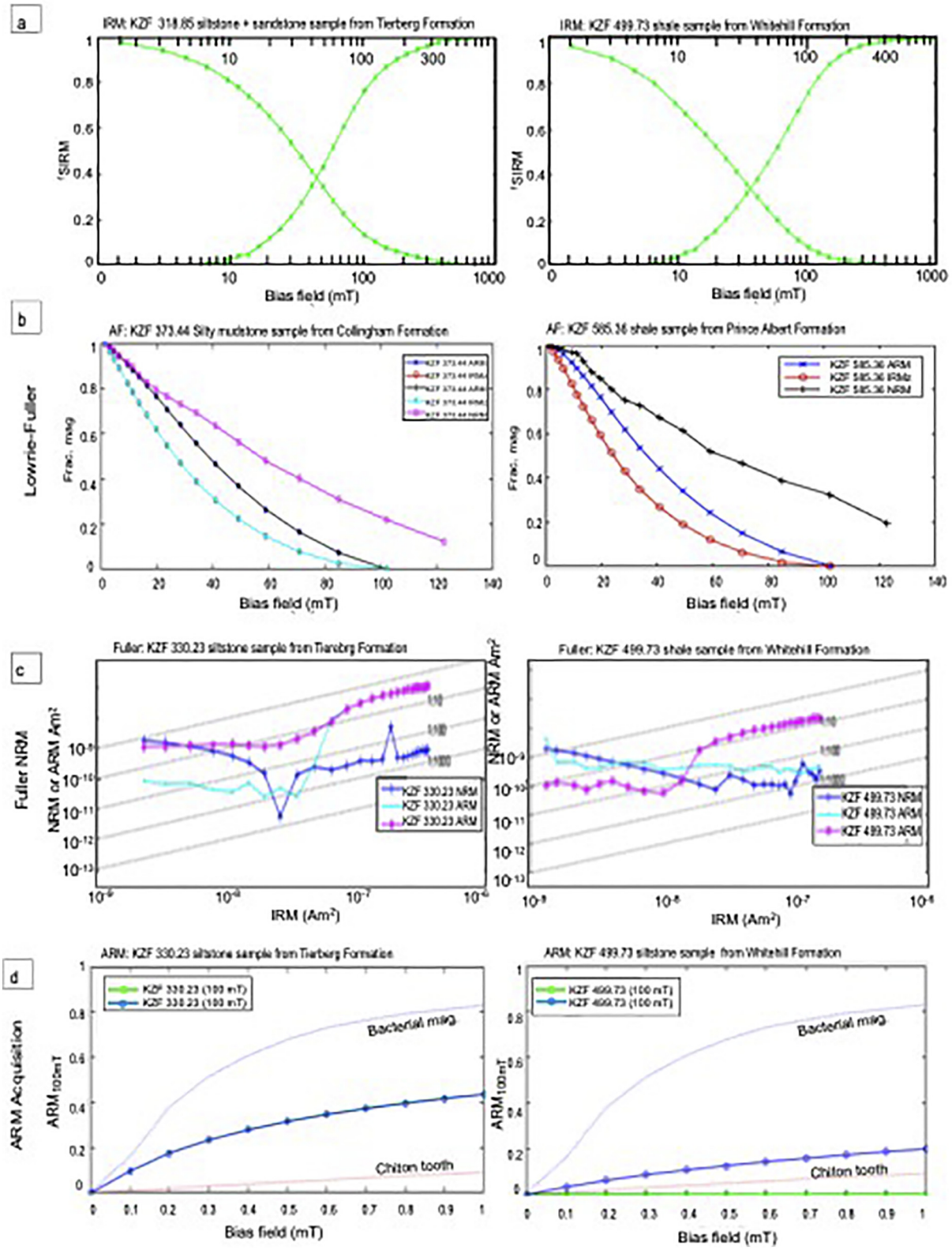


Fig. 5. a-d Rock magnetic results for siltstone/silty mudstone samples (all on the left) from the Tierberg/Collingham formations and shale samples (all on the right) from the Whitehill/Prince Albert formations as described in the text. All results are consistent with fine-grained pyrrhotite with a contribution from interacting single domain magnetite holding the ChRM.

origin are presented in Fig. 6e which suggests the syngenetic formation of these magnetic minerals early in the sedimentation process. The presence of graded beds in KZF-1 (Fig. 6e) suggests high sediment discharge and associated debris flows and transitional flows (Boulesteix et al., 2019); which is conducive for the preservation of magnetic carriers (Roberts and Turner, 1993).

Results from geochemical studies of phosphatic concretions from the Lowermost layer of the Prince Albert Formation by Herbert and

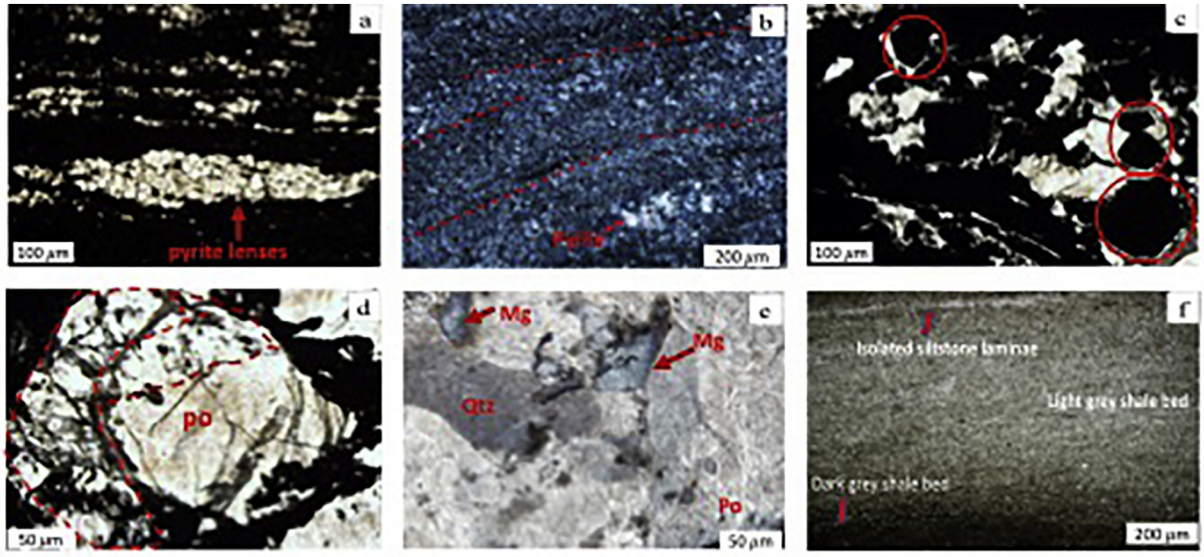


Fig. 6. Photomicrograph under reflected light (uncrossed polarizers): a) strips of syngenetic framboidal pyrite crystals along bedding surface in organic matter-rich black shale in laminae from the base of Whitehill Formation at 492.78 m depth; b) pyrite crystals in siltstone layer from the Collingham Formation at 366.76 m depth; c) bulk agglomeration of black-opaque euhedral magnetite grains (red circles) with inter-layered white - grey quartz from the lower portion of the Prince Albert Formation at 613.17 m depth; d) a pseudo-hexagonal pyrrhotite grain with visible growth of new mineral at the cleavage along cracks and grain boundaries (red outline) from the base of the Whitehill Formation at 499.66 m depth; e) a pyrrhotite grain with inclusions of elongated prismatic magnetite (red arrows) and quartz grain from the Prince Albert Formation at 578.96 m depth; f) graded bedding of siltstone laminae, dark grey to light grey massive shale bed from the Tierberg Formation at 60.34 m depth. Po: pyrrhotite; Qtz: quartz; Mg: magnetite.

Compton (2007) include $\delta^{18}O_{SMOW}$ values in the range of 7.8–8.9‰ which suggests precipitation in a mixture of meteoric and fresh glacial meltwaters; while $\delta^{13}C$ values ranging between −15 to −3‰ PDBh suggests increased microbial degradation of organic matter; and $^{87}Sr/^{86}Sr$ ratios ranging between 0.716 and 0.737, which is significantly greater than the mean ratio of 0.708 for Permian seawater (Veizer and Compston, 1974). The overall increased $^{87}Sr/^{86}Sr$ ratios in the lower Ecca Group rocks likely reflect the progressively increasing influx of continental crust-derived strontium which supplies the more radiogenic end-member component (Veizer, 1989) and indicates that the concretions have not been reset by later stage alteration or recrystallisation (Herbert and Compton, 2007). These interpretations support our petrography of possibly biogenic magnetite and preserved monoclinic pyrrhotite grains that could carry stable and primary magnetization in the Lower Ecca Group rocks.

4.3. Demagnetization

An NRM intensity of ~10^{-3} A/m was recorded for most of the shale samples, while the siltstone samples had a slightly weaker intensity of ~10^{-3} - 10^{-4} A/m (see Fig. 7). Progressive thermal demagnetization of the samples revealed three discernible components that were labelled as components A, B and C. Component A is recorded in ~70% of the samples and was removed by the AF and low-temperature demagnetization steps up to 150 °C. Component B was isolated by progressive thermal demagnetization up to 280 °C in most of the samples and 520 °C in a few B components were identified as very noisy. Overall, both A and B components did not display any identifiable groupings. A characteristic remanence (or C component) was identified in most samples after the removal of A and B components. Component C was mostly isolated as arc trajectories of demagnetization to which planes could be fitted. In some cases, C components were also identified as linear trajectories of demagnetization towards the origin (Fig. 7). Mostly, the C components were identified in an intermediate range of thermal demagnetization steps (i.e., 320–340 °C) (Fig. 7a) in the weaker magnetized samples. C components had a higher unblocking range of 480–580 °C (Fig. 7c and d) in the slightly more magnetized samples. The C components are dominantly steeply downward directed (+), with only a few samples being steeply upward-directed (−). The C+ direction (declination = 151.2°; inclination = 76.6°; $n = 72$; $R = 63.1$; $k = 7.97$; $\alpha_{95} = 6.33°$) is statistically antipodal from the C- direction (declination = 325°; inclination = −69.4°; $n = 18$; $R = 16.7$; $k = 13.44$; $\alpha_{95} = 9.79°$). The two directions pass a reversal test with a C classification. A 1.62° angle between the averages (λ_o) is much smaller than the statistical test value or critical angle (λ_c) of 13.29° at which the two polarity modes would be statistically distinguishable (McFadden and McElhinny, 1990). (See Fig. 8.)

5. Interpretation of results

5.1. Timing of remanence acquisition

Our coercivity spectra analysis and composite IRM experiments (Figs. 5a-d) shows that the main carrier of the ChRM in the studied section is a moderately interacting fine-grained single domain (SD) pyrrhotite and magnetite. Typically, pyrrhotite is believed to be the product of transformed pyrite crystals either by oxidation process to form clustering of magnetite around the pyrite crystals and subsequent sulphidation of the newly formed magnetite to pyrrhotite; or formed directly from pyrite through desulphurization (Li and Zhang, 2005). Since the KZF-1 core is from deep stratigraphic levels in the southwestern part of the main Karoo Basin, such effects of burial are plausible, but the implication of this would be the conversion of early diagenetic pyrite in the carbonaceous shale and mud of the lower Ecca Group strata to pyrrhotite during early burial alteration.

However, based on the petrographic analysis, there is no evidence showing the pyrite framboids or magnetite crystals as a precursor to the ubiquitous recrystallization or growth of pyrrhotite in the lower Ecca Group rocks in KZF-1. In addition, geochemical and thermodynamic studies suggest that formation of pyrrhotite is kinetically hindered during early burial in sedimentary environments based on its typically low temperature (Schoonen and Barnes, 1991, 1999; Lennie et al., 1995). The window for pyrrhotite formation typically starts at ~200 °C at a depth of ~6 km; and at temperatures up to 300 °C, it is typical for only neoformed pyrrhotite to carry the NRM (Rochette, 1987;

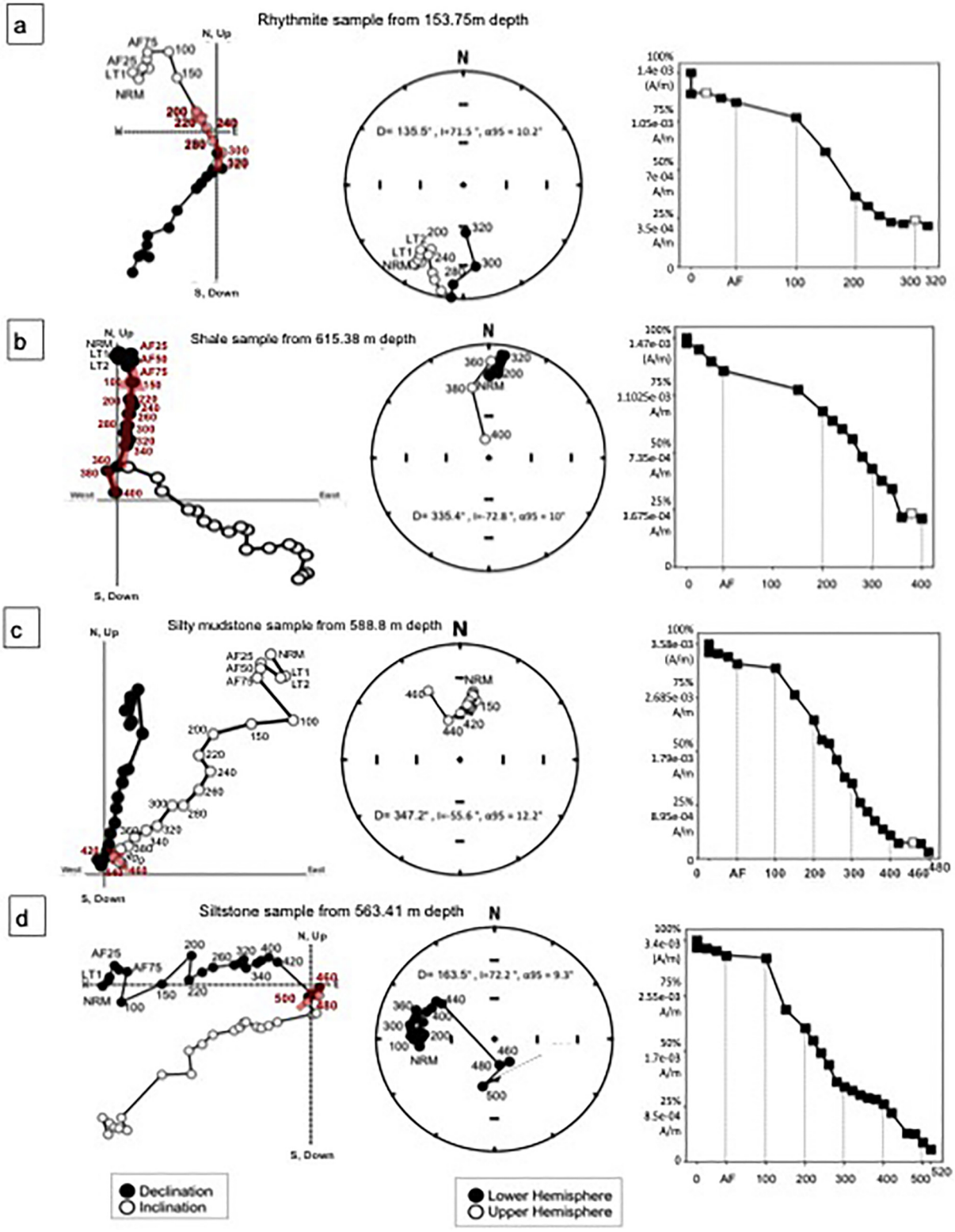


Fig. 7. Representative Zijderveld demagnetization diagrams and corresponding equal-area projections and J/Jo plots for the component C fits for samples from KZF-1. Note: a) removal of lower stability components before isolation of the ChRM (C+) as a plane fit from 200 to 320 °C; b): removal of lower stability components before isolation ChRM (C-) as a plane fit from AF75–400 °C; c): removal of lower stability components before isolation ChRM (C-) as a line fit from 420 to 460 °C; d): removal of lower stability components before isolation ChRM (C+) as a line fit from 460 to 500 °C. ChRM (component C) for each sample is highlighted in red; AF represents alternating field demagnetization steps in 10 mT; LT represent low-temperature measurement steps and numbers represent thermal demagnetization steps in °C. D = Declination values; I = Inclination values and α_{95}=radius of the cone of 95% confidence about the calculated mean directions.

(Dekkers, 1988; Dekkers et al., 1989). Reflection seismic and borehole data reveals that the deepest part of the Karoo Basin detected to be in the south, along the northern boundary of the Cape Fold Belt is only ~4 km in the southwest region (Scheiber-Enslin et al., 2015). Depth maps by Scheiber-Enslin et al. (2015) also show that the Whitehill Formation with abundant pyrite lenses and pyrrhotite nodules in this

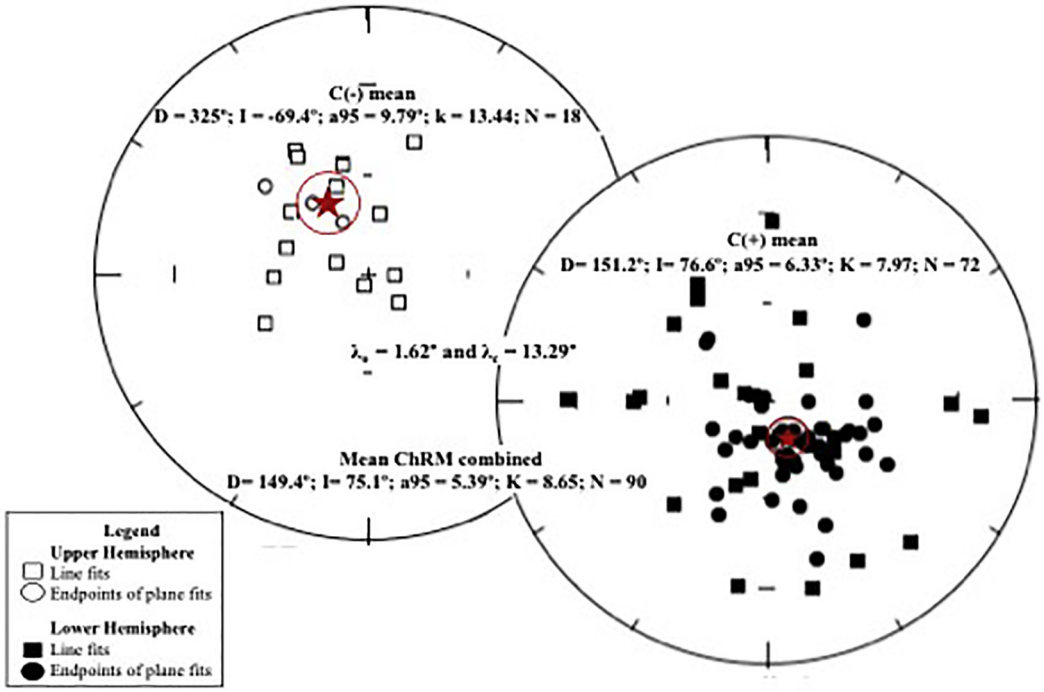


Fig. 8. Equal area plot displaying the combined mean reversed ChRM direction (C+) of the lower Ecca Group strata (on the top). The inset red oval and stars represents the cone of 95% confidence in the mean direction and the centre of the confidence circles respectively. The mean normal (C-); mean reversed (C+) ChRM direction; the angle between the means (λ_o) and the statistical test value or critical angle (λ_c) at which the two polarity modes would be statistically distinguishable are indicated.

region of the basin only reached a maximum depth of ~3 to 4 km. In addition, based on refraction seismic profile for the Karoo Basin by the Southern Oil Exploration Corporation (Pty) Ltd. (popularly known as "SOEKOR"), Scheiber-Enslin et al. (2015) reveals that an evident uplift in the south-central region of the basin brought lower density Ecca Group shales (1800–2650 kg/m^3) closer to the surface by ~2 km or greater.

Furthermore, despite the regional elevation of paleotemperatures of up to 200 °C for the southwest Karoo (Maré et al., 2016), relatively low porosity and thermal conductivity have been detected for the lower Ecca Group shales and mudstones in this region (Jones, 2003; Maré et al., 2016). This temperature interval is consistent with the potential Pb-loss in the Karoo Basin (Belica et al., 2017), which should typically only occur in detrital zircons during low-temperature processes (Schoene, 2014). Thus, burial-heating could not have resulted in the formation of pyrrhotite or the acquisition of a younger normal-polarity in the lower Ecca Group.

The probable explanation is the formation of pyrite crystals in the basin as a paramagnetic product from intermediate ferrimagnetic pyrrhotite (Roberts and Turner, 1993). This occurs in the presence of high concentrations of hydrogen sulphide (H_2S) in rocks which is considered to inhibit direct precipitation of iron sulphides (such as pyrrhotite) other than pyrite and the effective reduction of reactive and soluble iron phases in a reducing and organic matter rich environment (Roberts and Turner, 1993; (Horng and Roberts, 2006)Soliman and Goresy, 2012). We, therefore, interpret the presence of pyrrhotite in the mud rich lower Ecca Group rocks as traces of preserved remains of an intermediate sulphide phase resulting from an arrested pyritization process caused by the consumption of available hydrogen sulphide in low permeability mud successions before full reaction and formation of pyrite could occur (Roberts and Turner, 1993).

Restriction of pyritization is conducive to the formation of stable magnetizations in the mud rich lower Ecca Group rocks, and the co-existence of small-sized biogenic magnetite and preserved pyrrhotite grains (see Fig. 6e) suggests the syngenetic formation of these magnetic minerals in a relatively short period in the sedimentation process. Single domain biogenic magnetite grains produced by magnetotactic bacteria are considered a major carrier of magnetic remanence in deep water sediments (Kirschvink and Chang, 1984), and are most favourable for retention of magnetic remanence (Opdyke and Channell, 1996). Preserved pyrrhotite crystals, on the other hand, can serve as stable and strong magnetic remanence carriers in mud and shale, especially if enclosed within graded beds associated with high sedimentation rates (Roberts and Turner, 1993). Similar results of stable remanence magnetization in fine-grained sediments by co-existing biogenic magnetite and pyrrhotite were obtained by Mary et al. (1993) in Pliocene sediments from the Stirone River in northern Italy; and Roberts and Turner (1993) in late Miocene to early Pliocene age marine sediments from the north-eastern South Island, New Zealand.

The temperature-dependence behaviour of the NRM in KZF-1 is characterized by up to three components, namely: Component A – a low-coercivity and thermally unstable magnetization; component B which is a low-temperature component (200–280 °C); and component C which is an intermediate temperature (in most cases) component carried by the pyrrhotite population and/or high temperature (in a few cases) components carried by the magnetite population. Both components A and B did not define clear groupings and are not further discussed. We interpret the component C as a primary remanent magnetization acquired during deposition of the lower Ecca Group strata. Hence, despite being weakly magnetized, the lower Ecca Group strata possess a primary remanent magnetization. Our paleomagnetic results comprising of ChRMs with both normal and reverse polarities in the lower Ecca Group rocks is supported by a positive reversal test with a C classification. A corresponding paleopole was calculated as being situated at −54.9° S, 44.3° E, dp/dm = 9/9.8 and α_{95} = 9.11°.

5.2. Early Permian paleopole and pole comparison

To further constrain the relative timing of the interpreted primary magnetizations, the mean declination, inclination, associated α_{95} and site location (Lat. S, Long. E) were used to calculate the virtual geomagnetic pole (VGP) for the lower Ecca Group strata in KZF-1. This

calculated mean paleopole from the KZF-1 ChRM is plotted alongside reported Permian poles for the Karoo Basin (Belica et al., 2017; Tohver et al., 2015; Lanci et al., 2013; De Kock and Kirschvink, 2004). Our calculated paleopole for the combined ChRM data set from the lower Ecca Group strata reported in Table 2 is similar to the Permian pole positions reported for the middle-upper Ecca Group (Belica et al., 2017), lower Beaufort Group (Lanci et al., 2013; Tohver et al., 2015), and rocks that potentially span the PTB (De Kock and Kirschvink, 2004). Our KZF-1 pole is placed at a medium to high paleolatitude of −54.9° and is in good accordance with other Permian poles in the Karoo despite the age difference and data set from sampling sites spread over areas higher up in the stratigraphic levels of the Ecca Group strata. However, our KZF-1 pole for the lower Ecca Group strata shows an elongated VGP distribution that is coincidental with the 290 Ma segment of the APWP as opposed to a circular and symmetric VGP distribution that is typically produced by unshallowed inclinations (from a GAD) (Domeier et al., 2011).

It is often necessary to correct the paleomagnetic results obtained from sedimentary rocks for inclination shallowing caused by burial compaction, since the magnetization may acquire a shallow inclination bias as much as the order 10–20°, particularly for mud-rich marine sediments (Kodama, 2012). These inclination errors in sediments result in a paleolatitude error of up to 20° and an anomaly resembling that caused by contamination of the dipole field by a zonal octupole term (Rochette and Vandamme, 2001). The methods used for correcting for inclination shallowing are the DRM tensor method (Jackson et al., 1991); the Elongation Inclination (*E*-I) method (Tauxe and Kent, 1984) using the statistical field model TK03.GAD on a large data set (>100); the E-I method (Tauxe and Kent, 2004); and a correction method that considers inclination shallowing as a tangent–tangent relationship, hence its correction is carried out based on assumed flattening factors (f) using an inclination correction equation by King (1955):

$$\tan\ (\text{observed inclination}) = f \times \tan\ (\text{inclination at deposition}).$$

$$\tan\ (I_O) = f\ \tan\ (I_f).$$

where *f* is the shallowing amount represented as an empirically derived flattening factor (Tauxe et al., 2008) with typical values between 0.54 and 0.79 and a mean value of 0.65 for magnetite dominated rocks (Bilardello and Kodama, 2010). Since it is difficult to quantify the DRM tensor, and since our dataset is less than 100 to use the either of the E-I methods, we adopted the inclination shallowing correction method by King (1955) and applied a flattening factor of 0.6 on the ChRM of KZF-1.

The depositional process that is biased towards the horizontal and elongate/equant mineral grains are evidence that the lower Ecca Group rocks may have undergone some degree of compaction resulting in inclination shallowing. However, the already steep inclination of the majority of the ChRM, suggests a bias of varying magnitude due to grain settling and subsequent compaction flattening processes. Thus, we first corrected the DRM using a factor of 0.6 on all inclination value and obtained a corrected inclination of 80.5 with a flattening of 5.4 to the previous inclination value of 75.1° and a 95% confidence circle (A_{95}) of 6.33°. To avoid over-correction for the DRM, we applied a fixed 45° cut-off for shallowing bias corrections as shallowing has less effect on steep inclinations that are already close to horizontal. This yielded a corrected inclination of 75.9° with a flattening of 0.8° and a 95% confidence circle (A_{95}) of 4.93° (Fig. 9 and Table 2).

The corrected inclination was used to recalculate the pole for the lower Ecca Group strata, which was then plotted with Permian poles reported for the Karoo Basin (De Kock and Kirschvink, 2004; Lanci et al., 2013; Tohver et al., 2015; Belica et al., 2017), and the Gondwana pole Apparent Polar Wander Path (APWP) by Torsvik et al. (2012) calculated using the spherical spline method, with applied corrections factor of 0.6; a moderate smoothing factor (500) and input poles weighted by their Q-factor (Van der Voo reliability criteria). The uncorrected and corrected KZF-1 paleopoles using an applied 45° cut-off angle shows good accordance with the published Karoo Permian poles and are consistent with the 290 Ma segment of the APWP. The corrected pole with applied factor (f = 0.6) on the DRM, on the other hand, does not overlap with the Karoo Permian poles or the 290 Ma segment (Fig. 9) of the Gondwana APWP of Torsvik et al. (2012).

To avoid any bias towards over-correction of the DRM, the 45° cut-off is recommended in this study and adopted for further discussions. The mid-palaeolatitude calculated for the lower Ecca Group rocks especially after applying the shallowing correction is consistent with the 290 Ma segment and predicted deflection curve of the Gondwana APWP. Also worthy of consideration is the low k values obtained, which is consistent with a low level of spurious effects in the palaeomagnetic results and serves as a useful measure of the homogeneous nature of the magnetization throughout the core. This is also supported by the consistent MAD values (Fig. 7) obtained from fitting of the primary components. Therefore, we do not attribute the relative position of our pole or slight dispersion of our paleomagnetic data to experimental errors, but rather to the prevalence of widespread shallow inclination bias in the poles derived from the paleomagnetic records of Early Permian sedimentary rocks by Torsvik et al. (2012) and slightly weak remanent magnetization of the rocks respectively.

5.3. Magnetostratigraphy

The 90 samples recording the ChRM define discrete stratigraphically constrained polarity groupings. These were used to develop a magnetostratigraphic profile for KZF-1. Negative inclinations (C-) are here taken as representing normal polarity, while positive inclinations

Table 2
Mean paleomagnetic directions of KZF-1. The calculated VGP position calculated using the Fisher mean is compared with selected Permian poles for the Karoo Basin.

Section	N(n)	Dec. (°)	Inc. (°)	K	α_{95}°	P	dp/dm	Site location		VGP		A95°	Reference
								Lat °S	Lon °E	Lat °S	Lon °E		
Component C - ChRM Normal	18	325.4	−70.4	12.82	10.04	N		−32.8	19.7				**This study**
Component C + ChRM Reversed	72	151.2	76.6	7.97	6.33	R							
Mean ChRM Combined	90(64.5)	149.6	75.3	8.62	5.4	R	9/9.9			−54.7	43.7	9.16	
Mean ChRM (45° cut-off applied)	**90(64.5)**	**149.7**	**76.1**	**10.09**	**4.94**	**R**	**8.4/9.1**			**−53.8**	**42.0**	**8.39**	
Mean ChRM corrected (f = 0.6)	90(64.5)	149.6	80.6	16.8	3.75	R	6.9/7.2			−47.9	33.4	6.37	
Upper Ecca Group	226 (199.5)	117.3	62.8	7.27	3.98	R	4.9/6.2	−32.7	20.06	−40.8	77.4	5.5	Belica et al. (2017)
Lower Beaufort (Abrahamskraal)	234		−62.9		9.72	N	8.6/11	−32.7	20.6	−56.1	74.1	9.73	Tohver et al. (2015)
Lower Beaufort (Ouberg pass)	136	308	−62		8.5	N	10.2/13.2	−32.4	20.3	−48	80	11.6	Lanci et al. (2013)
PTB	35	312	−62	17.8	7.6	R-N-R	9.1/11.8	−32.1	26.0	−51	86	10.4	De Kock and Kirschvink (2004)

Notes: K = best-estimate of the precision parameter for observed distribution of site-mean VGPs (Fisher, 1953); α_{95}=radius of the cone of 95% confidence about the calculated mean direction; N = number of samples used for calculation of the mean direction, n = L + 1/2 p (L = lines; p = planes; McFadden and McElhinny, 1988); P = Polarity; dp/dm = semi-minor/semi-major axes of oval of 95% confidence; A_{95} = the radius of the 95% confidence circle about the calculated mean pole. Recommended pole for this study is in bold. The VGP latitude and longitude values for the recommended pole for this study is in bold.

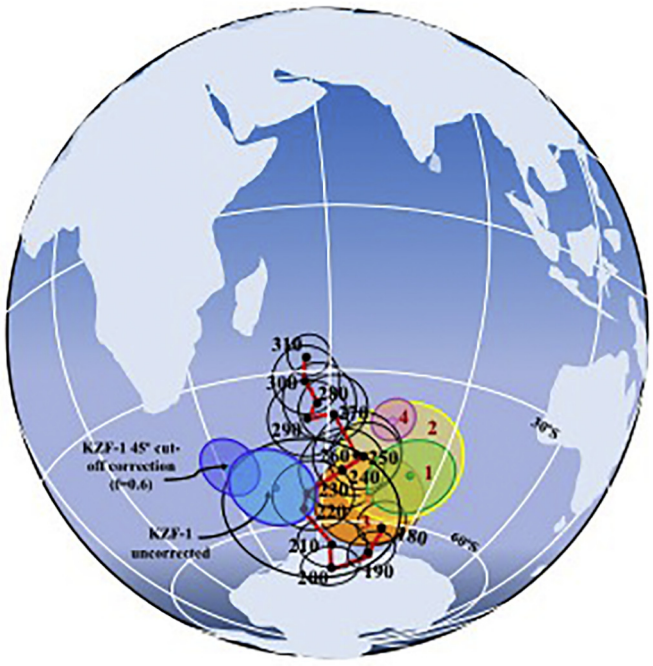


Fig. 9. The calculated pole for ChRM components of the lower Ecca Group strata in KZF-1 plotted alongside the Permian poles for the Karoo Basin and the recommended Gondwana APWP (red lines) for South Africa by Torsvik et al. (2012) spline path corrected for inclination shallowing using a flattening factor (f) of 0.6 and smoothing factor of 500. Compared Permian poles are 1: PTB (De Kock and Kirschvink, 2004); 2: Ouberg Pa ss (Lanci et al., 2013); 3: Abrahamskraal Formation (Tohver et al., 2015); 4: Middle-Upper Ecca Group (Belica et al., 2017). The calculated KZF-1 paleopole shown are uncorrected, correction applied only on inclinations values that are <45° and corrected using f = 0.6 for all ChRM.

(C+) are taken to indicate reverse polarities. Two or more consecutive samples of the same polarity are used to define a magnetozone. Zones of unidentifiable polarity due to poor sample coverage or the lacking identification of primary magnetizations are designated as grey bars.

The top of our section begins within the bottom ~350 m of the Tierberg Formation and underlies previously published composite section from the OR1 core (intersecting the top ~400 m of Tierberg and overlying Skoorsteenberg formations) drilled in the Tanqua depocenter (Belica et al., 2017). Occasional single samples recording normal polarity (C-) are observed scattered throughout our section (Fig. 10). However, these magnetozones constrained by the behaviour of single samples are regarded with some level of uncertainty in our reversal stratigraphy, as most of them are located close to zones of unknown polarity due to insufficient sampling. Four zones of unknown polarity were recorded in our section, namely: twice within the Tierberg Formation at depths of 51–78 m depth and 189–219 m depth; at depth of 432–467 m depth recognised as the thrust zone where duplication of the Whitehill Formation is observed due to thrusting of the underlying Prince Albert Formation; and lastly, at 631–655 m depth at the base of the Prince Albert Formation.

Our data also revealed dominantly reversed polarity intervals with three short normal subchrons recorded by shale and mudstone samples with a steep upward direction (Fig. 12). These were observed as normal subchrons recorded in the Tierberg Formation at ~31–51 m depth; at the base of the uppermost layer of the Collingham Formation at ~356.36–370.7 m depth; and at ~615.38–622.82 m depth within the Prince Albert Formation.

Demagnetization of the first of the two normal bearing sample (at 356.36 m-depth) that defined the brief normal subchron at the uppermost layer of the Collingham Formation revealed an A component attributed to viscous remanence and removed by 240 °C, and a B component unblocked at relatively at an intermediate temperature around 380 °C. Both of these two components are restricted to normal polarity and the ChRM component is isolated at a higher temperature of 440 °C with a distinct convergence (endpoint of) that is anchored to the origin (see Fig. 10). The second negative polarity bearing sample (at 370.7 m-depth) showed similar behaviour to the underlying one with the A component at 100 °C, a B component removed at 280 °C and the ChRM component isolated at 340 °C through a line fit of an endpoint of three steps that are anchored to the origin (see Fig. 10).

The stable remanent magnetisation components for the three samples that defined the normal subchron at the base of Prince Albert Formation were determined from Zijderveld demagnetization plots and equal area vector projections in at least four successive measurement steps between 380 and 580 °C that indicate a well-defined component of ChRM constrained by linear decay towards the origin for two of the samples and a plane fit for the third one (red highlighted vector in Fig. 11) after removal of the soft magnetization components A and B.

The normal polarity samples from 615.75–622.82 m-depth from the lowermost portion of the Prince Albert Formation (see the supplementary data) were closely sampled due to a noticeable change in lithology from siltstone to silty mudstone to shale. This closely spaced sampling interval (i.e., four samples within 7 m) shows pervasive normal polarity rocks characterized by the removal of a low coercivity, low-temperature component followed by a northerly down or southerly up intermediate temperature component that tended towards a line missing the origin, and the last component which we interpret as the primary component that demagnetized to normal and constrained by line and plane fits tied to origin (see Fig. 11).

Our interpretation using plane fits is not restricted to a single great circle on its own given that it provides inadequate information to estimate the direction of the hidden component; so importantly, with the generation of more than one great circle from different specimens, these great circles converge and allows for a better-constrained estimate of directions. In addition, several independently available direct estimates obtained through stable endpoints of other specimens in the section, are also used to access the information available in the numerous great circle. In this regard, the reliability of our results obtained from the arc method is verified by directly assessing the estimates independently available from other specimens isolated through line fits or stable endpoints in our section. We discovered that the assigned polarity from the plane fits is consistent between the several great circles and direct observations determined by linear segments of stable endpoints.

Alternative interpretations of the characteristic directions of these samples might be to align the declination of the low-temperature, low-coercivity component (parallel) to the present field declination, and infer counter-clockwise rotation to validate the normal polarity magnetization. However, it is worthy to note that application of this technique is most applicable to sites that have near-equatorial paleolatitudes and uncertain northern vs. southern hemisphere positions through a time where it is inappropriate to assign polarity zones based solely on inclination patterns, which in this case does not apply to the Karoo Basin. Also, this alternative interpretations of the characteristic directions samples in our section more or less assumes that the overprint vector of present-day north-directed polarity is often superimposed on the primary polarity vector and completely ignores that the last three to four demagnetization steps of these samples (i.e., third component) which is isolated at higher temperatures, and show a clear convergence (endpoint) to the origin (see Figs.10 and 11).

6. Discussion

6.1. Stability and origin of the NRM

A combination of positive $\delta^{18}O_{SMOW}$ and negative $\delta^{13}C$ values for the lower Ecca group rocks (Herbert and Compton, 2007) suggests colder temperature due to presence of ice sheets, upwelling, increased erosion

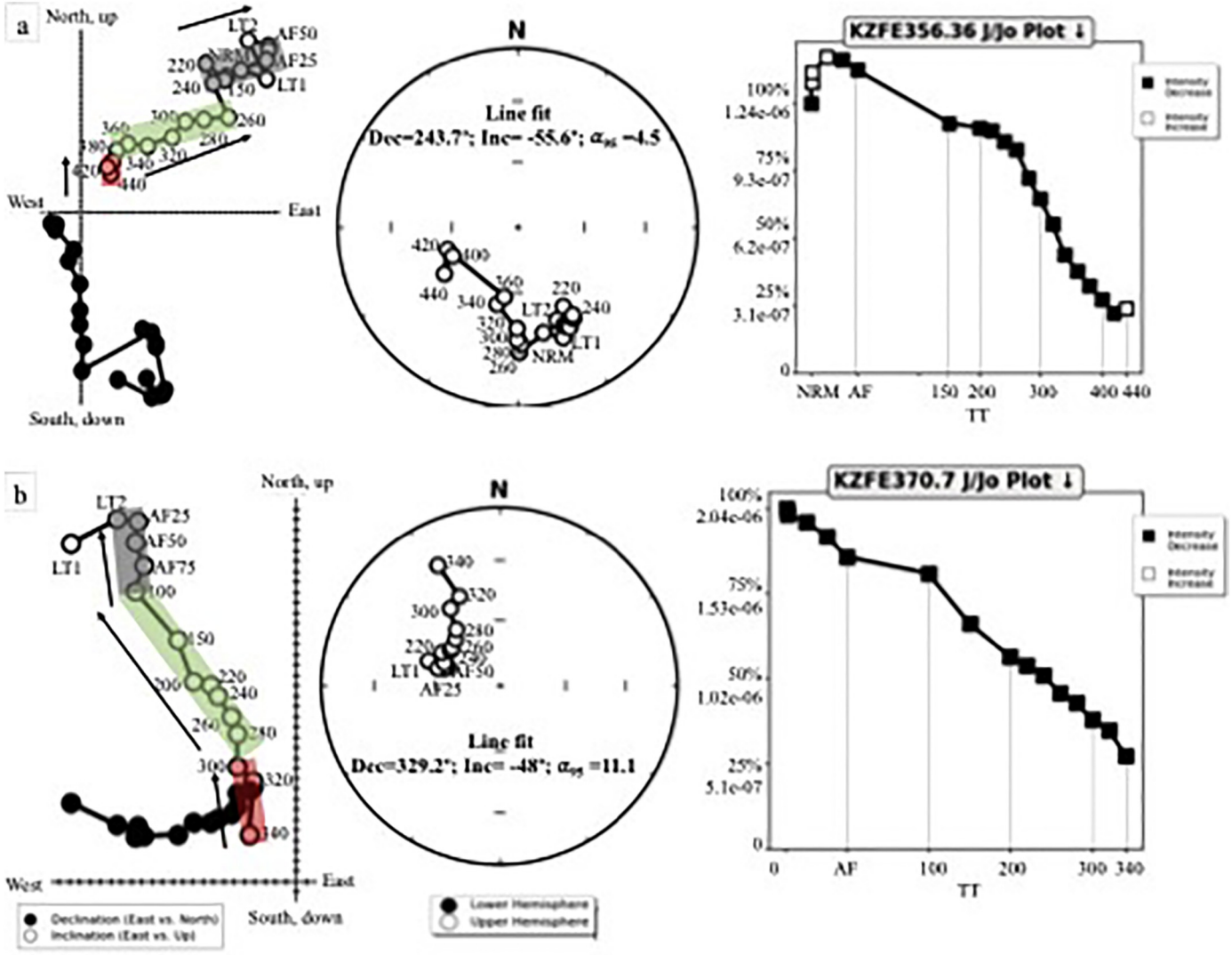


Fig. 10. Representative Zijderveld demagnetization diagrams, corresponding equal-area projections and J/Jo plots showing the A (grey highlight), B (green highlight) and C components (red highlights) line fits for the samples from the uppermost layer of the Collingham Formation. Note: a) removal of lower stability components before isolation of the ChRM (C+) as a line fit from 400 to 440 °C; b): removal of lower stability components before isolation ChRM (C-) as a line fit from 300 to 340 °C; AF represents alternating field demagnetization steps in 10 mT; LT represent low-temperature measurement steps and numbers represent thermal demagnetization steps in °C. Dec = Declination values; Inc. = Inclination values and α 95=radius of the cone of 95% confidence about the calculated mean directions.

from land that carry detrital iron minerals and intense volcanism resulting in increased emission of CO_2 being buried as organic carbon which is essential for the formation and preservation of bacterial magnetite. The uppermost and lowermost section of KZF-1 shows magnetite as the ChRM carrier (see processed data in appendix A). This is expected given that the dissolved oxygen restricted to the sediment surface (i.e., a few centimetres) is rapidly used by bacteria to produce CO_2 from reactive organic matter. Also, at greater depth, progressive consumption of reactive organic matter and the breakdown of detrital iron minerals to soluble ferrous ions by bacterial magnetosomes as well as inorganic reactions (which is necessary for pyrite formation) occurs in anoxic environments (Roberts and Turner, 1993)(Roberts et al., 2011). Thus, the high organic matter content and an anoxic environment in the Whitehill and Prince Albert formation accounts for the preservation of fine-grained bacterial magnetite towards the base of the cores.

The entire middle section of the core consisting of lighter-coloured, finely laminated siltstone and fine-grained sandstones with mud laminae in the middle-lower portion of the Tierberg Formation and the clay and tuff rich strata in the Collingham Formation is dominated by pyrrhotite-bearing rocks. These strata were deposited in the distal part of the Ecca interior seaway and are composed of a neritic and diatomaceous zone reflecting strong upwelling and a condensed section with abundant carbonate concretions and ash beds (see Section 2 and Fig. 3f-g). The laminated silt and fine sands suggest high sedimentation rate and the ash layers which typically alter to clay (i.e. tuffaceous layers or bentonite) contributed ash-derived Nutrients that increased the biological productivity (confirmed by observed burrowing in Fig. 3f) and preservation of organic carbon in the rocks (Minisini et al., 2017). This scenario facilitated a hydrogen sulphide rich environment and arrested pyritization process and a resultant dominance of preserved pyrrhotite grains in this section of the core.

The post-Ordovician zircon grains of 282–281 Ma and 275–274 Ma obtained from the lower Ecca Group rocks has been linked to two volcanic pulses (Fildani et al., 2009) from a continental magmatic arc source beyond the CFB i.e. the early margin volcanic activity in the Choiyoi volcanic province, Northern Patagonia (McLachlan and Jonker, 1990; Veevers et al., 1994; Viljoen, 1994; McKay et al., 2015). The constrained ages of the zircon grains correspond with the ages of the two normal magnetozones in our magnetostratigraphic section (see Fig. 12). Thus, the U–Pb age data for detrital zircons in the lower Ecca Group rocks

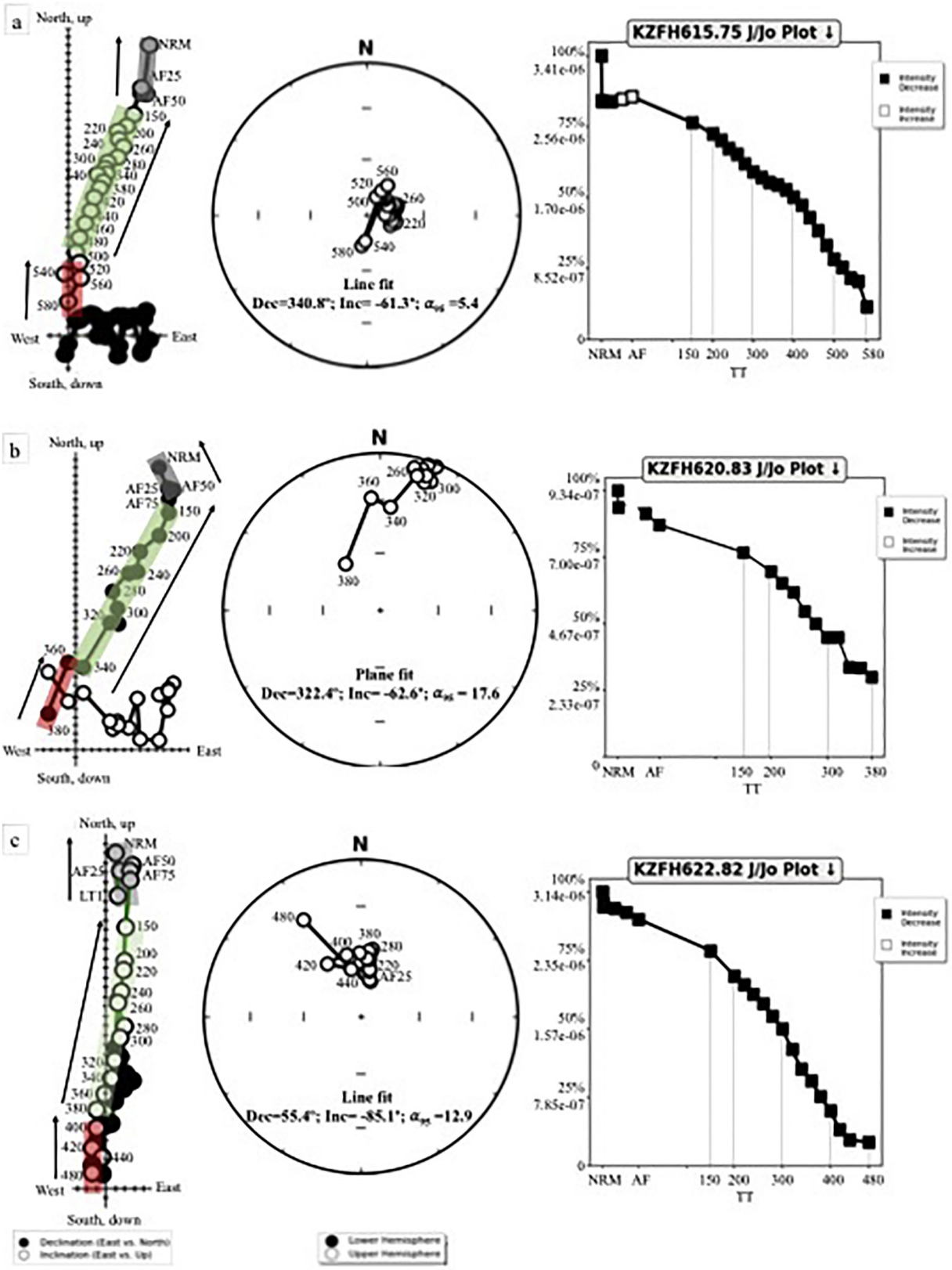


Fig. 11. Representative Zijderveld demagnetization diagrams, corresponding equal-area projections and J/Jo plots showing the A (grey highlight), B (green highlight) and C components (red highlights) line fits for the samples from the lower layer of the Prince Albert Formation. Note: a) removal of lower stability components before isolation of the ChRM (C+) as a line fit from 540 to 580 °C; b): removal of lower stability components before isolation ChRM (C-) as a plane fit from AF75–380 °C; c) removal of lower stability components before isolation of the ChRM (C+) as a line fit from 400 to 480 °C. AF = Alternating field demagnetization steps in 10 mT; LT = low-temperature measurement steps and numbers represent thermal demagnetization steps in °C. Dec = Declination values; Inc. = Inclination values and α 95=radius of the cone of 95% confidence about the calculated mean directions.

not only serves for correlation but also reflects the proto-source of the zircon grains, which is represented in KZF-1 by pervasive tuffaceous and volcanic ash beds sourced from an eruptive event. Coupled with a high supply of fine-grained sediments in a reducing environment, it accounts for the syngenetic formation of the magnetic carriers in the core.

The possible contribution of thermal remanent magnetization (TRM) by pyrrhotite at the detriment of magnetite (Butler, 1992) during thermal demagnetization was prevented by the passing of a gentle flow of nitrogen gas as a blocking measure to prevent oxidation of ferromagnetic minerals (see Section 3.3). Furthermore, at thermal demagnetization up to 340 °C and above, the samples did not show a significant change in remanence directions but rather displayed a consistent direction with those obtained during AF demagnetization (see Fig. 7 and supplementary data in appendix A). Such consistency indicates the negligible effect thermal alteration has on the NRM of magnetite and pyrrhotite holding the primary magnetization.

The site location of the KZF-1 borehole is situated below the southern limit of the Karoo Jurassic dolerite sills and dykes, hence the absence of dolerite intrusion in the core. Previous studies by Ballard et al. (1986) and Lanci et al. (2013) conducted on the lower Beaufort and upper Ecca group rocks respectively suggest remagnetization of the rocks in these investigated sites. The study by Ballard et al. (1986) showed widespread remagnetization at least for Late Permian and Early Triassic time based on a negative fold test and the closeness of the obtained paleopole for these rocks to the Karoo igneous and intrusive Jurassic APWP (Ballard et al., 1986). However, the investigated sites of both studies are ~2000 km above in the stratigraphy from the lower Ecca Group rocks.

Investigation of remagnetization of the lower Ecca Group rocks in the southwestern Karoo which is lower down in the stratigraphy (i.e., the Tierberg, Collingham, Prince Albert and Whitehill formations) conducted by Maré et al. (2014, 2016) proposes that the Karoo dolerite sill intrusions did not completely overprint the magnetic signatures of the Karoo sedimentary strata especially lower in the stratigraphy as was previously reported by Ballard et al. (1986). These studies by Maré et al. (2014, 2016) were based on anisotropy of low field magnetic susceptibility (AMS) data confirming that re-magnetization of the magnetic fabric only within distances that are half of the sill thicknesses. The magnetostratigraphy from these studies further shows that negative polarity similar to the dolerite paleomagnetic direction were only recorded in rocks within very short distances to the dolerite contact aureoles. Thus, the normal polarity ChRM could not have originated from remagnetization by the Karoo dolerites nor is it consistent with the present-day field or expected GAD.

The relative IRM and ARM values for the pyrrhotite and magnetite bearing rocks are roughly similar (see Fig. 5a-d) and less than 25% of magnetization in both rock types are lost at 100mT AF treatment (see Fig. 7). The positive reversal tests obtained in this study further show the ChRM to be primary, and the rock magnetic experiments indicate that the carriers of magnetization are PSD fine-grained ferrimagnetic (monoclinic) pyrrhotite and possibly bacterial magnetite, which are stable and capable of preserving the primary Early Permian remanence.

In addition, we believe that the scatter in our paleomagnetic data may be due to a few reasons such as the probable nature of rapid deposition of the lower Ecca Group sediments in the high-water content and

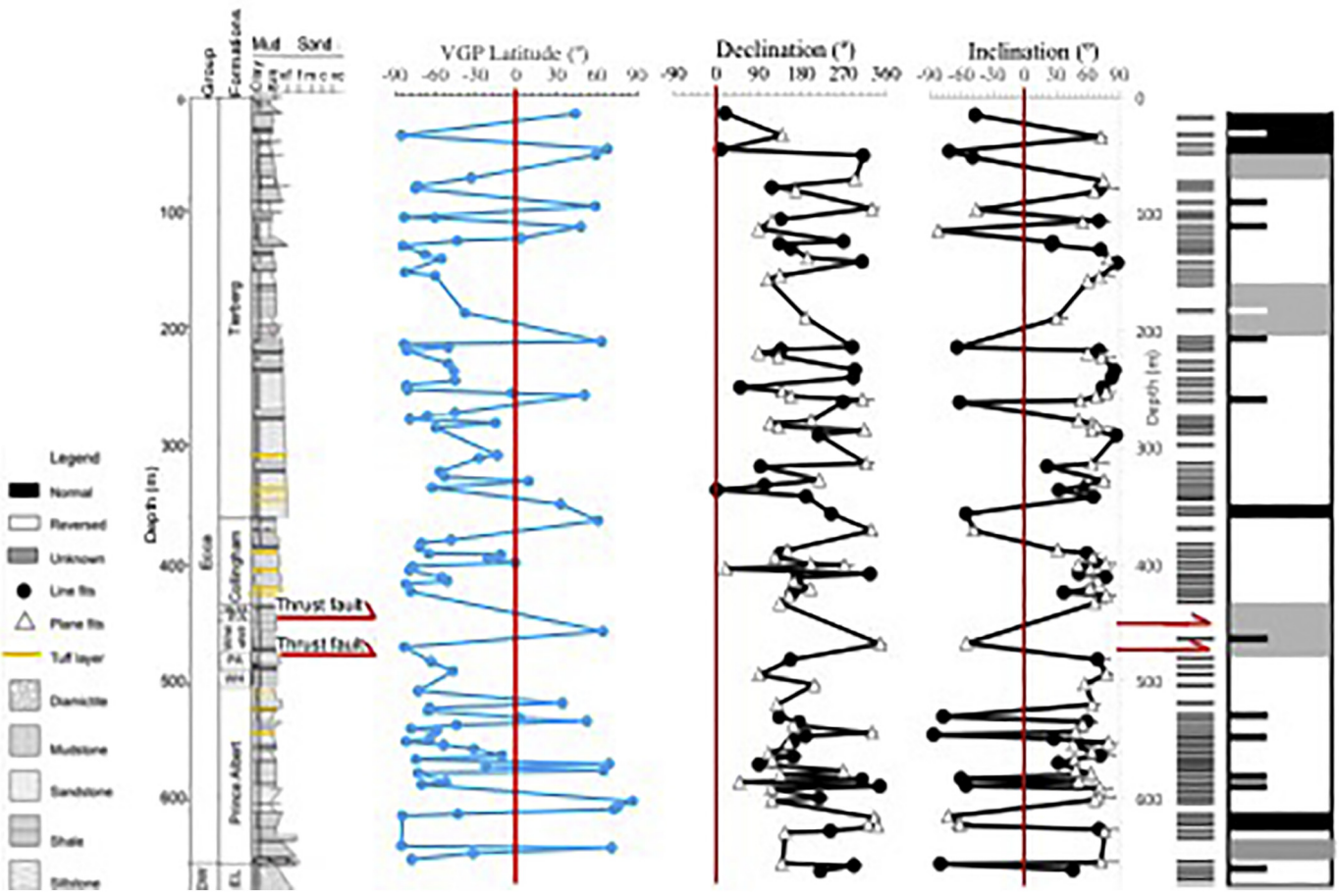


Fig. 12. ChRM directions and magnetostratigraphic results from the KZF-1 section. Sampled levels are indicated and paleomagnetic directions of samples that gave good ChRM are plotted against stratigraphic depth. Note: DW represents Dwyka Group and EL represents Elandsvlei Formation.

shelf edge setting, which eased the widespread positioning of the DRM (Kodama, 2012). Also, due to the weak nature of our rocks, it is possible that the resolution limits of the magnetometer may have contributed to the observed scatter in our data.

6.2. Early Permian magnetostratigraphic correlation

The relative age constraints for the lower Ecca Group strata of the south-western Karoo Basin in this study are based on correlation with the composite reference section proposed for the Early Permian by Hounslow and Balabanov (2018) and tied with published U–Pb SHRIMP and CA TIMS ages for the southwestern Karoo Basin. Three normal polarities (C-) subchrons are recorded in KZF-1 (Fig. 12). The first of these is recorded by three consecutive samples at the lower portion of the Prince Albert Formation. Informed by the CA TIMS (zircon) ages of 281.15 Ma + 0.72/−1.12 and 282.17 Ma + 0.32/−0.44 by Griffis et al. (2019) for single zircon grains from volcanic ash layers from the middle and lowermost portion of the Prince Albert Formation in the Laingsburg area in the southwest area of South Africa and correlative strata in the Paraná Basin of Brazil, we correlate the normal subchron at the lowermost portion of the Prince Albert Formation in our section to the Cl2n late Artinskian normal magnetozone on the section by Hounslow and Balabanov (2018) and thus assign it an age of 281 Ma (Fig. 13). The CA TIMS ages represent recently revisited volcanic ash layers from the Prince Albert Formation that yield better-constrained ages than that proposed by Bangert et al. (1999) and serve for correlation of deglaciation events across southwest Gondwana (i.e., South Africa and Brazil).

The second short normal magnetozone recorded by two samples at the topmost layer of the Collingham Formation is correlated to the Cl3n Kungurian normal magnetozone (Fig. 13) on the magnetostratigraphic compilation of Hounslow and Balabanov (2018). This correlation proposes a ~ 276 Ma age constraint for the topmost layer of the Collingham Formation. This age constraint is tied to U–Pb SHRIMP ages of 270 ± 1 Ma for the ash layers within the formation (Turner, 1999) and ~ 275 Ma proposed for a 3 cm-thick ash bed situated about 12 m from the base of the Collingham Formation (Fig. 13) in the Laingsburg depocenter (Fildani et al., 2007).

wBased on our magnetostratigraphic result, a Kungurian age is suggested for the Whitehill Formation underlying the Collingham Formation (Fig. 13). This corresponds fairly well to radiometric ages of 280.5 ± 2.1 Ma for the Whitehill Formation in the Aranos Basin of southern Namibia (Werner, 2006) and a 278.4 ± 2.2 Ma age for the stratigraphic-equivalent shales of the Irati Formation (Anderson and McLachlan, 1979) in the Paraná Basin of southern Brazil (Santos et al., 2006). Furthermore, this result potentially clarifies the debate of a Kungurian age (Cole and Barbolini, 2019) as opposed to an early Roadian age (Götz et al., 2018) for the carbonaceous shale formation (see Fig. 2). In addition, our findings suggest a late Kungurian to late Roadian age (~ 276–270 Ma) for the lower portion of the Tierberg Formation. This age corresponds, firstly, with a U–Pb SHRIMP age of 270.4 ± 2.7 Ma for zircon grains from the middle of the Tierberg Formation from the Laingsburg depocenter (McKay et al., 2016) located 50 km to the southeast (20.75°E), and secondly, with the 269.5 ± 1.5 Ma age reported for the topmost portion of the formation (Belica et al., 2017).

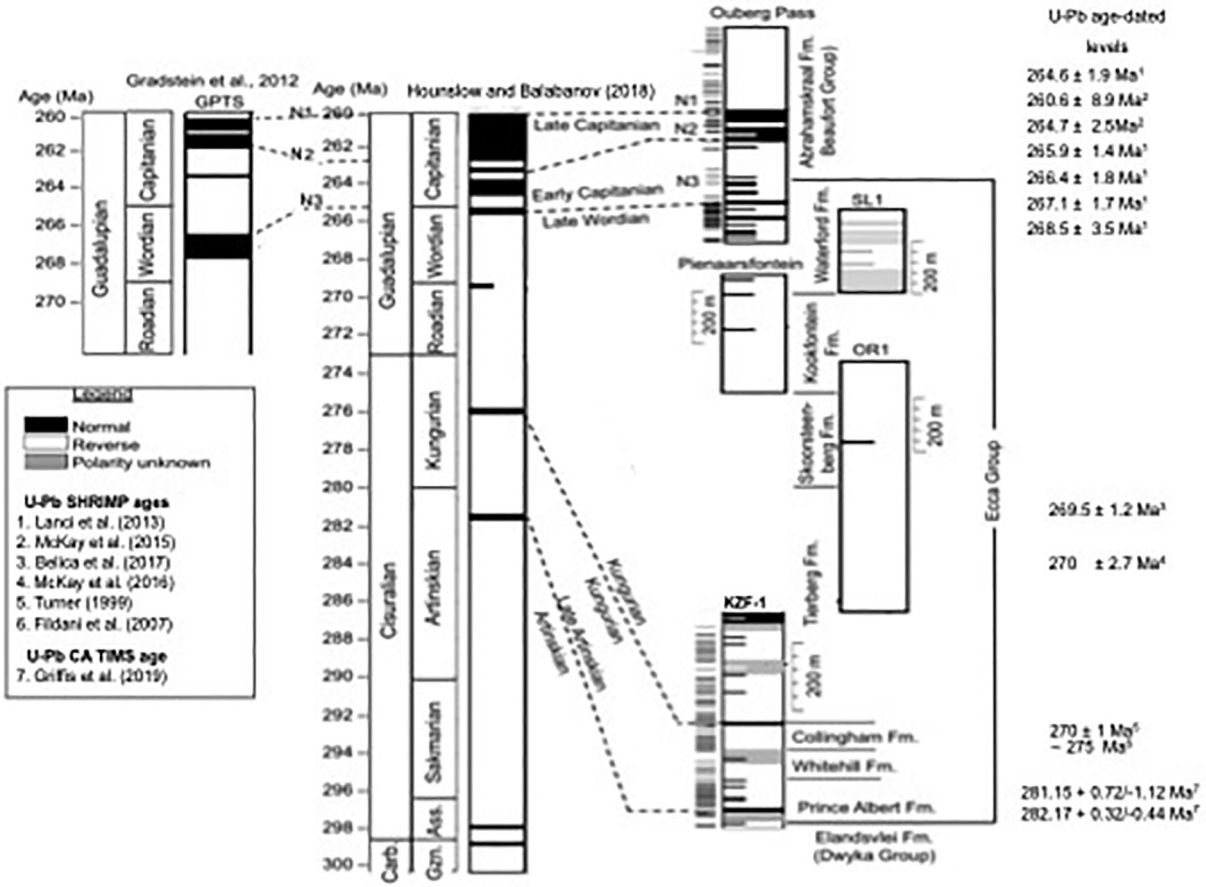


Fig. 13. Composite magnetostratigraphy of the Permian Ecca Group strata in the southwestern Karoo Basin of South Africa. KZF-1 section shows the magnetostratigraphy of the lower Ecca Group rocks and its correlation with the composite reference section proposed by Hounslow and Balabanov (2018) and U—Pb SHRIMP / CA TIMS age constraints for the southwest part of the Karoo Basin (far right).

Most importantly, our results re-establishes the overlying magnetostratigraphic sections for the middle-upper Ecca Group (Belica et al., 2017) that reports an Artinskian age for the uppermost part of the Tierberg Formation (see Fig. 1). Our magnetostratigraphic constraint suggests that the magnetostratigraphic correlation of a late Artinskian and mid-Kungurian age for the overlying Skoorsteenberg and Kookfontein formations respectively by Belica et al. (2017) is inexact (Fig. 2). Firstly, as stated earlier in Section 1.3, the magnetostratigraphic age constraints of 281–275 Ma for the middle-upper Ecca Group proposed by Belica et al. (2017) conflicts with their U–Pb SHRIMP age of 269.5 ± 1.2 for the underlying Tierberg Formation, as it suggests older relative ages for the overlying Skoorsteenberg and Kookfontein formations. Secondly, the Skoorsteenberg Formation consists of a thick succession of five deep-water sand-rich submarine fan complexes (Bouma and Wickens, 1991; Wickens, 1994; Bouma, 1997), and the Kookfontein and Waterford formations consist of deltaic strata suggested to have been rapidly deposited by large prograding deltas (Visser, 1993; Wickens, 1994; Wild, 2005); thus, a short and rapid sedimentation period of ~7 million years is expected (as suggested in Fig. 13).

Furthermore, based on the existing U–Pb dated levels and the magnetostratigraphic constraints (correlated to the GPTS proposed by Gradstein et al., 2012) obtained for the lower Beaufort Group (Lanci et al., 2013) and the lower Ecca Group strata in this study (Fig. 13), the middle to Upper Ecca Group is likely Wordian in age. Overall, our magnetostratigraphic age constraints of Artinskian (281 Ma) to Kungurian (276 Ma) for the lower Ecca Group strata is in agreement with isotopic ages and biostratigraphic constraints from the overlying strata.

7. Conclusions

Despite being weakly magnetized, our results show that the lower Ecca Group strata record a primary remanent magnetization acquired by monoclinic pyrrhotite and fine-grained biogenic magnetite. This conclusion is supported by a positive reversals test combined with the relative thermal stability of the remanence. Furthermore, there is generally in good accordance with the reversal stratigraphy with other Karoo Permian results. Our paleomagnetic pole calculated for the lower Ecca Group in the southwestern Karoo Basin scores 5/7 on the Van der Voo criteria of paleomagnetic reliability (Van der Voo, 1990), and our composite magnetostratigraphy for the whole of the Ecca Group in the southwestern Karoo Basin represents a potentially valuable tool towards evaluating the correlation and temporal development of the Ecca Group throughout the Karoo Basin.

Our results present an improvement of the Early Permian chronostratigraphy in the southwestern Karoo Basin using a magnetostratigraphic approach supplemented by rock magnetic and petrographic studies. The rock magnetic analysis and paleomagnetic measurements closely agree, and as for samples in the upper section, the ChRM showed an intermediate unblocking temperature of 320 °C – 340 °C with a steep up inclination and interpreted as a component residing in pyrrhotite; while samples in the lower part of the section showed a maximum decay of 480° C – 580 °C with a steep down inclination and interpreted as a component residing in possibly biogenic magnetite.

The observed magnetic polarity units in KZF-1 exhibit correlation with the composite magnetostratigraphic section proposed for Permian rocks and published U–Pb SHRIMP and CA TIMS ages. Our dominantly reversed polarity punctuated by short normal subchrons confirms that deposition of the lower Ecca Group strata occurred during the Kiaman reverse superchron (KRS) i.e., ca. 318 to 265 Ma (Opdyke et al., 2000; Steiner, 2006; Hounslow and Balabanov, 2018). Hence, we propose a depositional age constraint of ~281–273 Ma for the Early Permian lower Ecca Group strata in the southern Karoo Basin. Our findings better constrain the chronostratigraphy of the southwestern Karoo Basin, and therefore, could serve as a global correlation tool for Permian successions and allows for a paleogeographic reconstruction of the depositional environments concerning the Ecca interior seaway in the Karoo Basin.

CRediT author statement

Abosede Abubakre: Conceptualization, Methodology, Software. Abosede Abubakre: Data curation, Writing-Original draft preparation. Abosede Abubakre: Visualization, Investigation. Michiel De Kock: Supervision. Michiel De Kock: Software, Validation. Michiel De Kock: Writing- Reviewing and Editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.gr.2020.11.009.

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